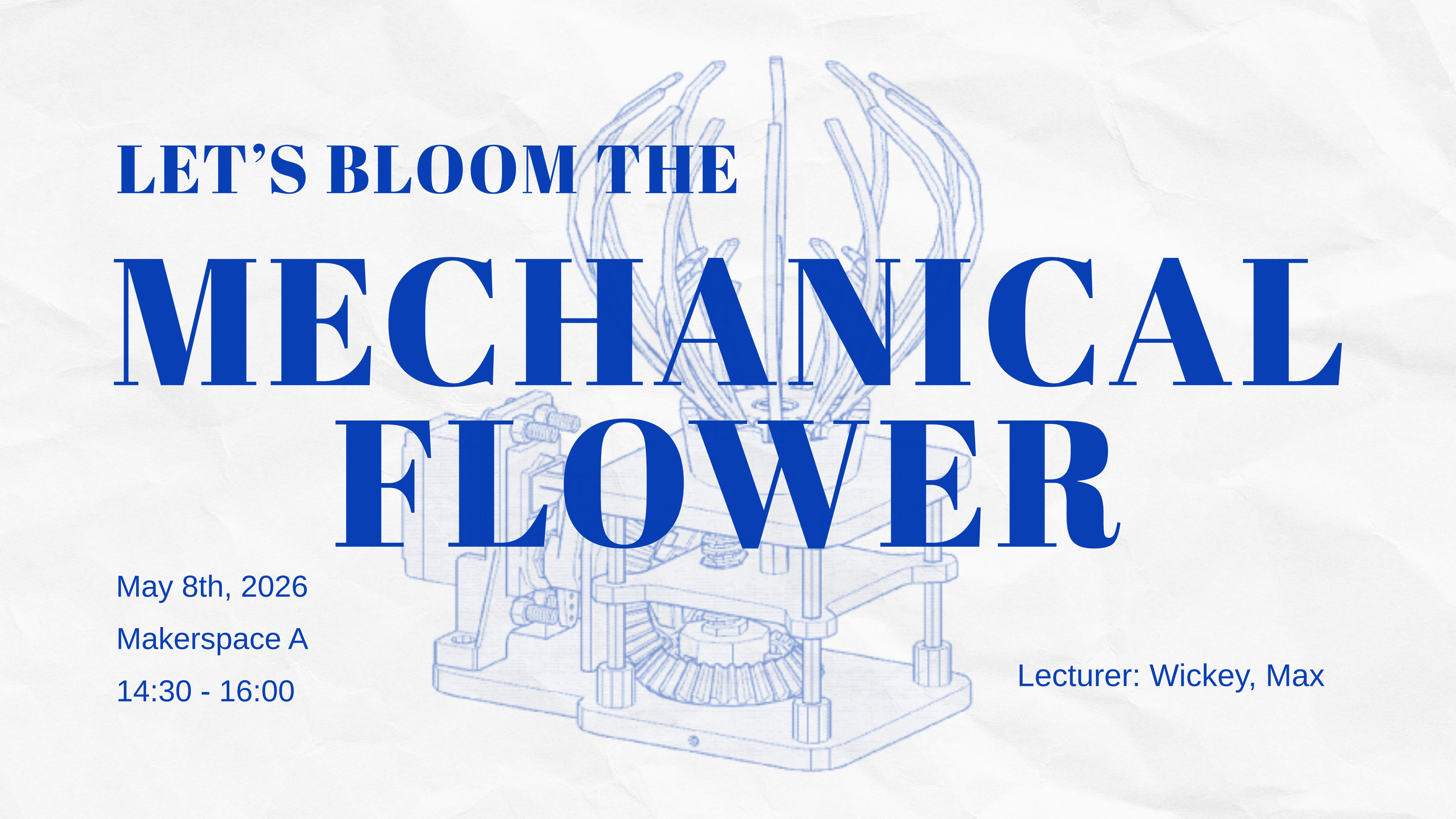




LET'S BLOOM THE MECHANICAL FLOWER

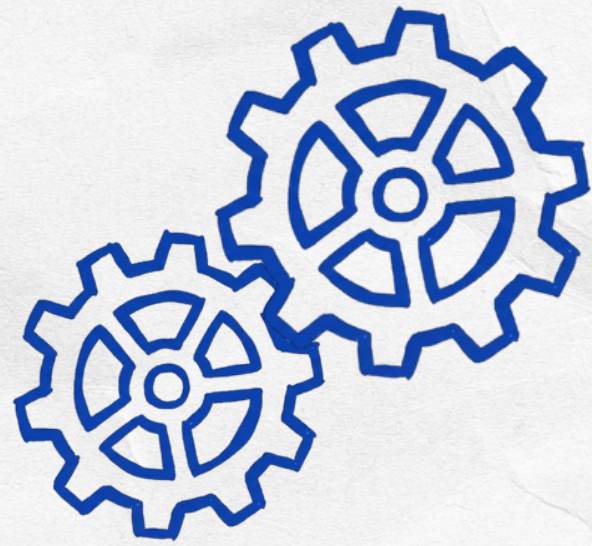
May 8th, 2026

Makerspace A

14:30 - 16:00

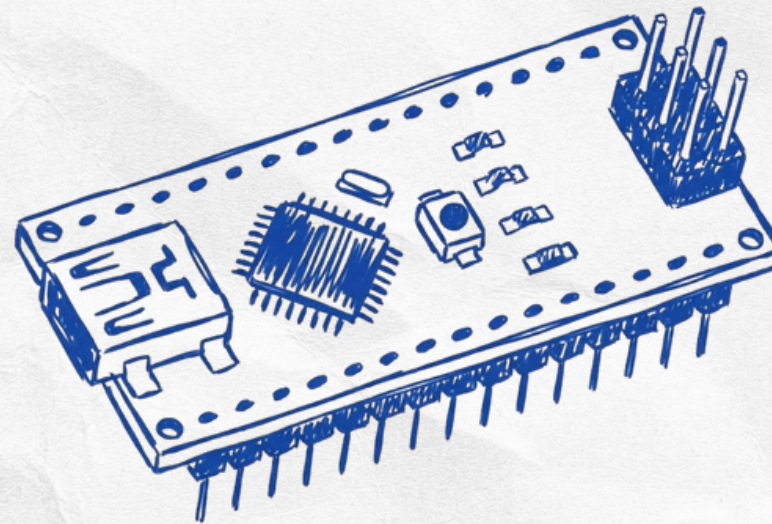
Lecturer: Wickey, Max

introduction



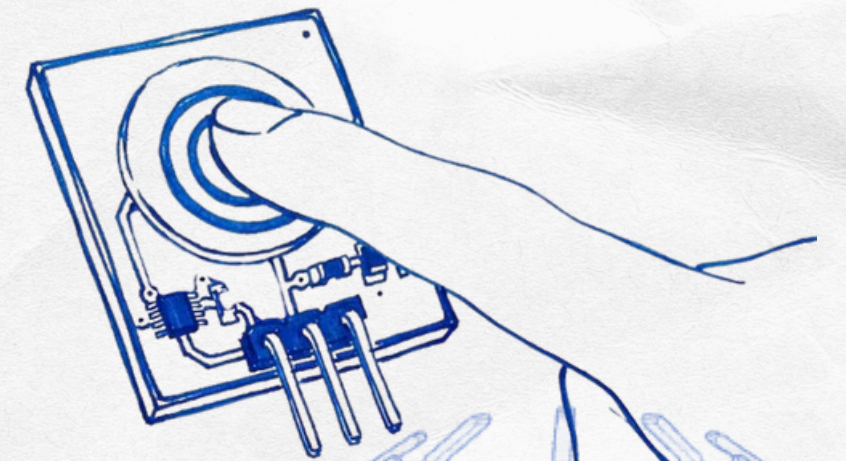
Transmission

How gears drive the flower bloom?
Step-by-step to assemble it!

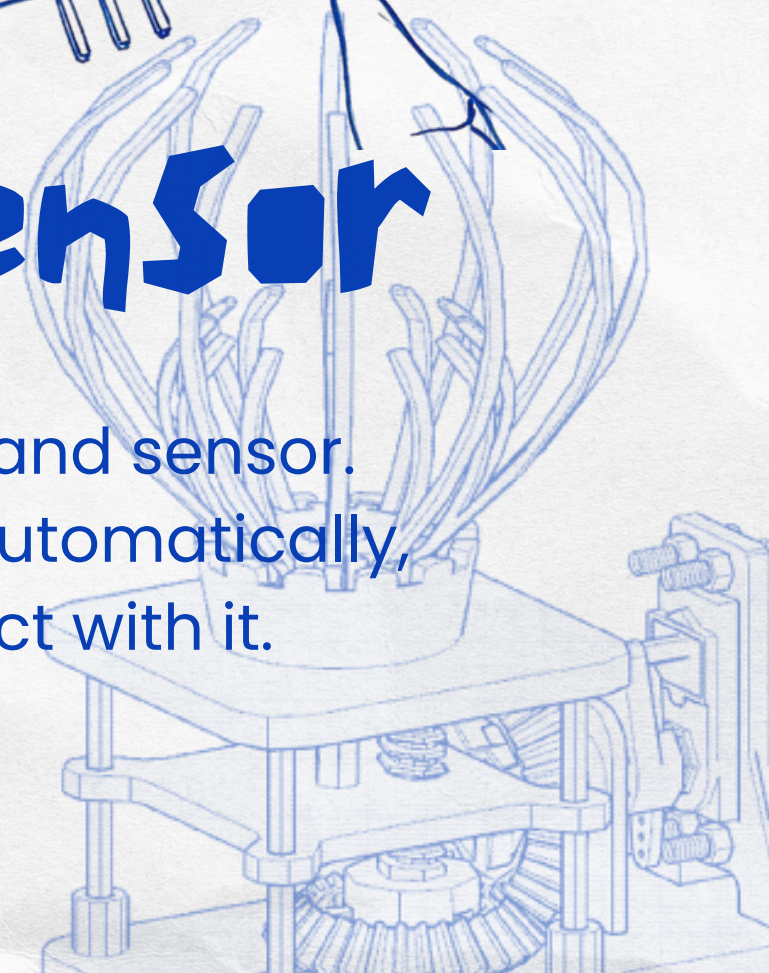


Arduino

Connect the Arduino Nano with servo and sensor.
Then your flower will respond to touch automatically,
blooming and closing as you interact with it.



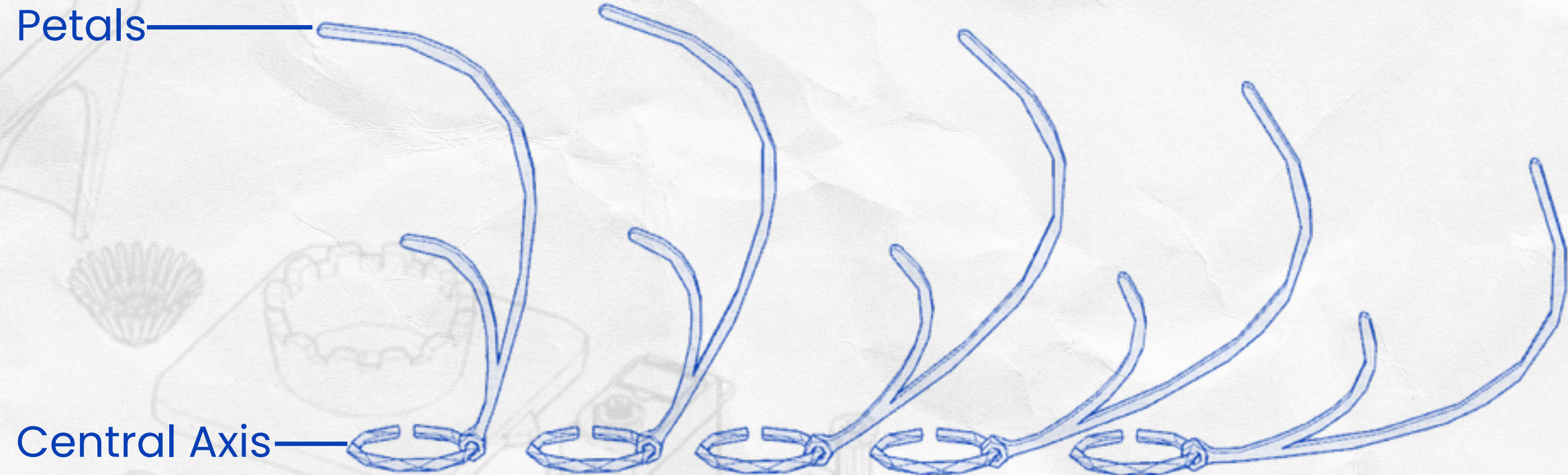
Sensor



How do the mechanical parts move like flower?

How a real flower in nature blooms?

1. Let the petal parts move like a real flower

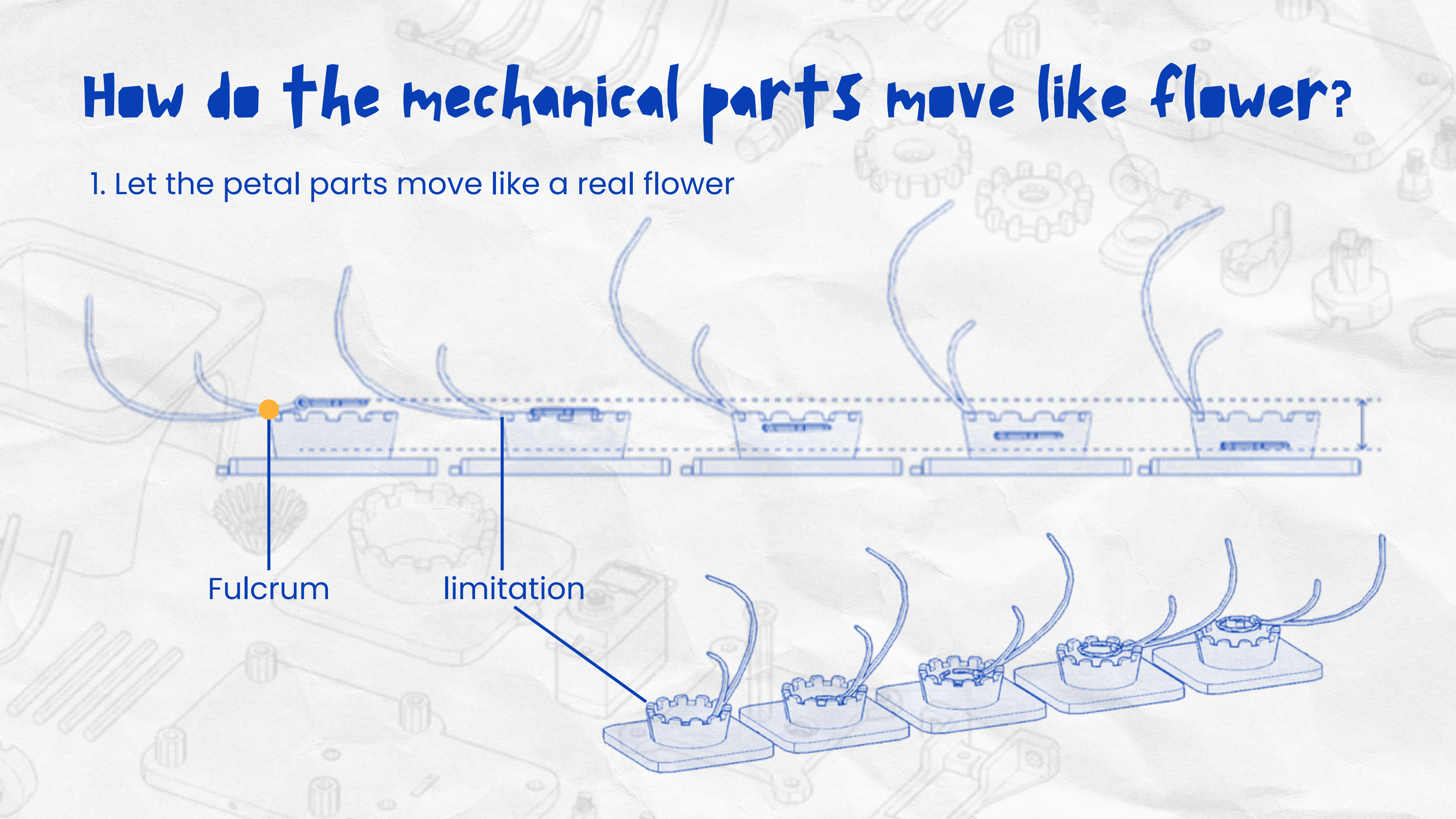


How do the mechanical parts move like flower?

1. Let the petal parts move like a real flower

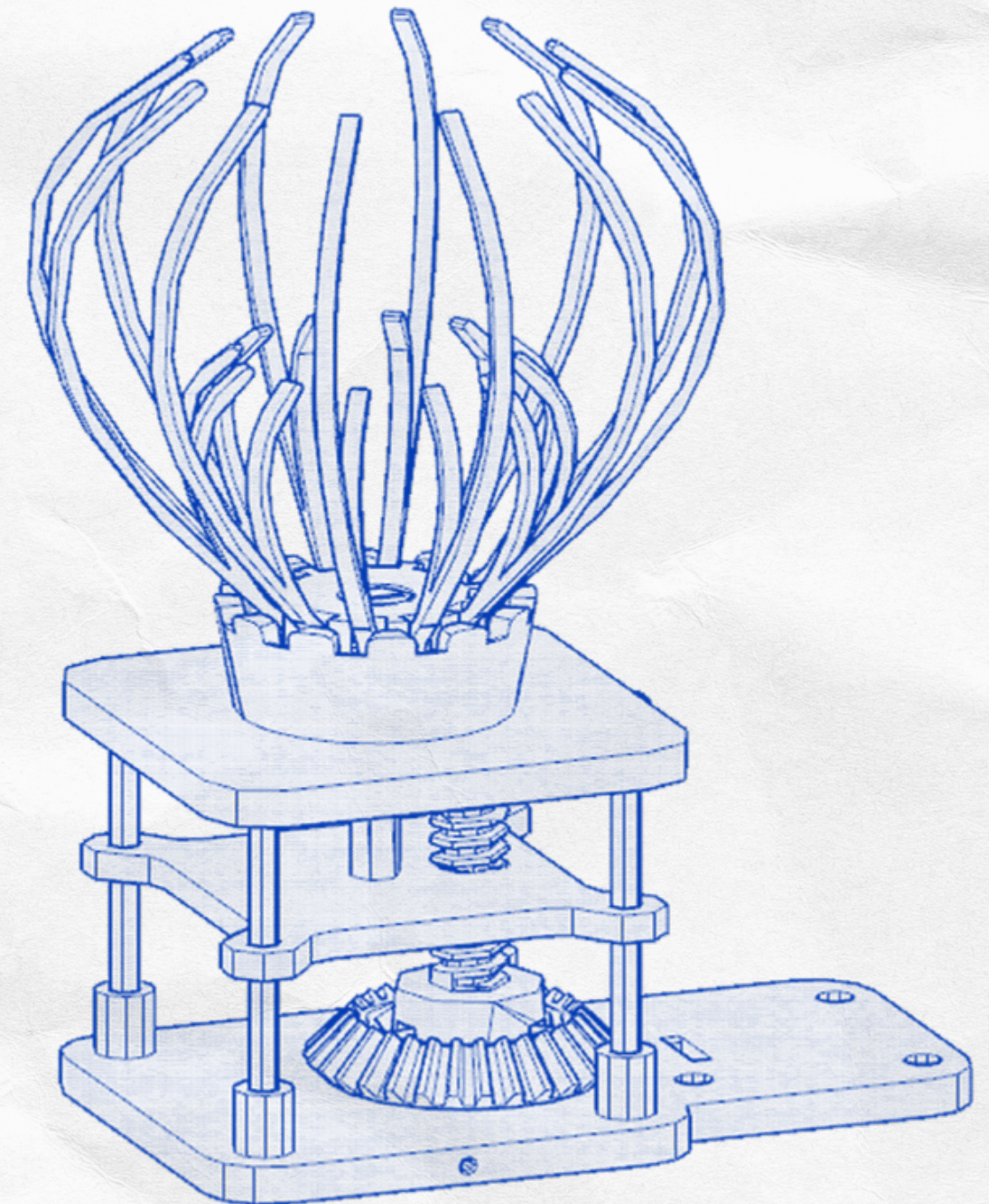
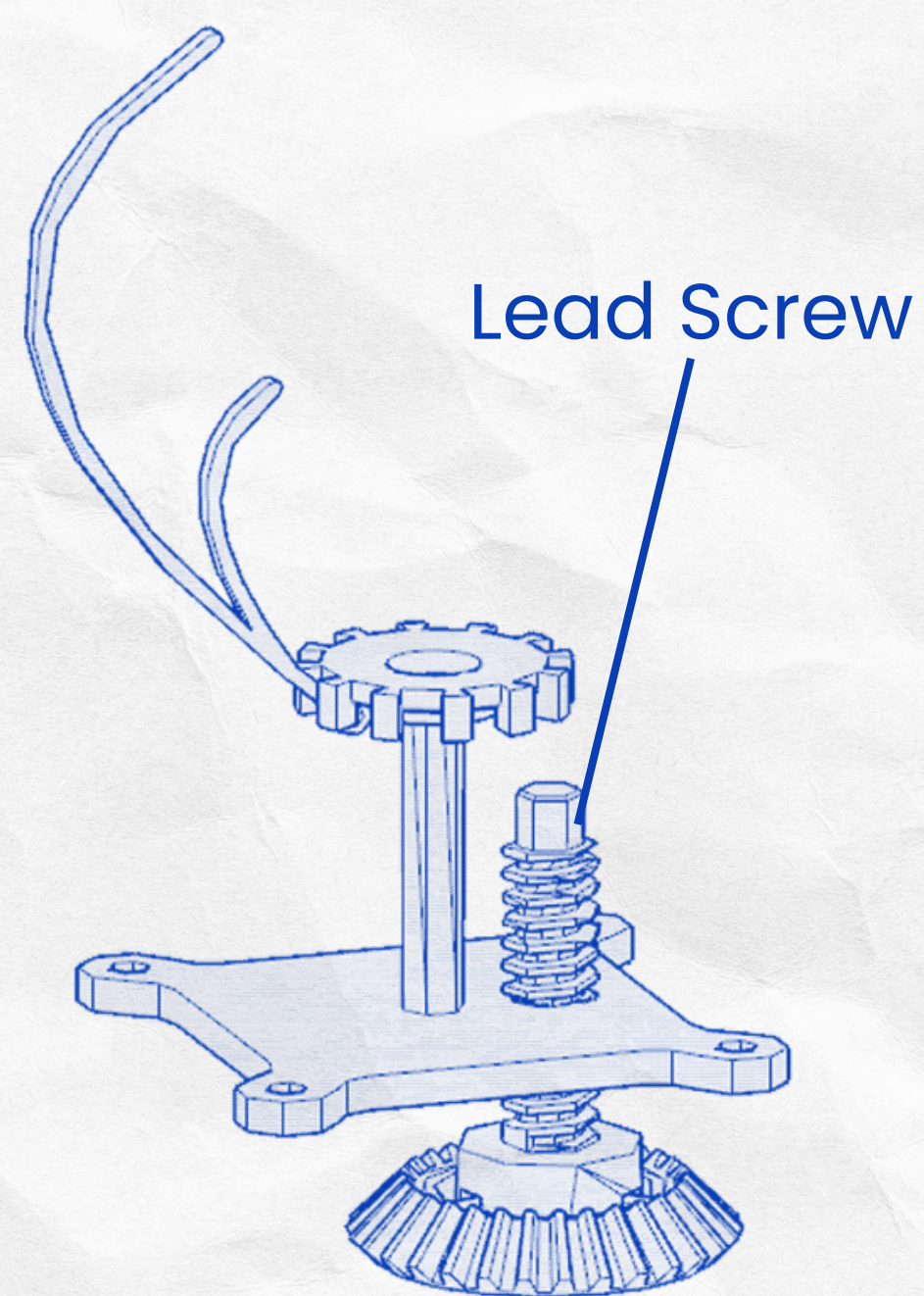
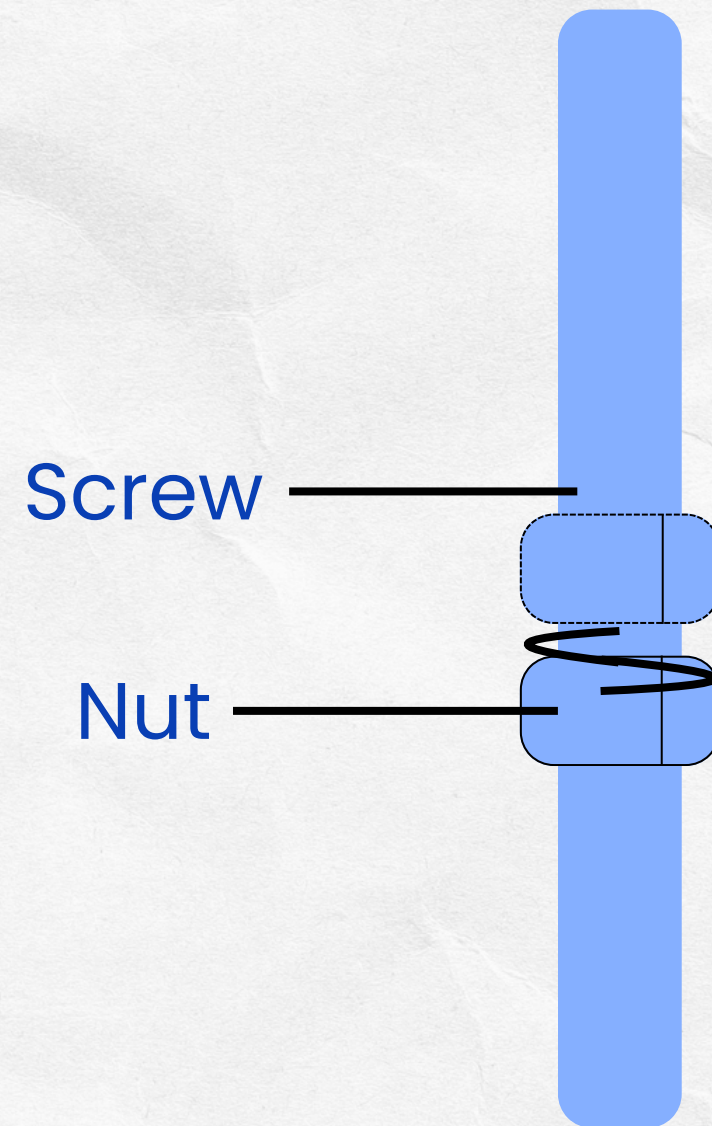
Fulcrum

limitation



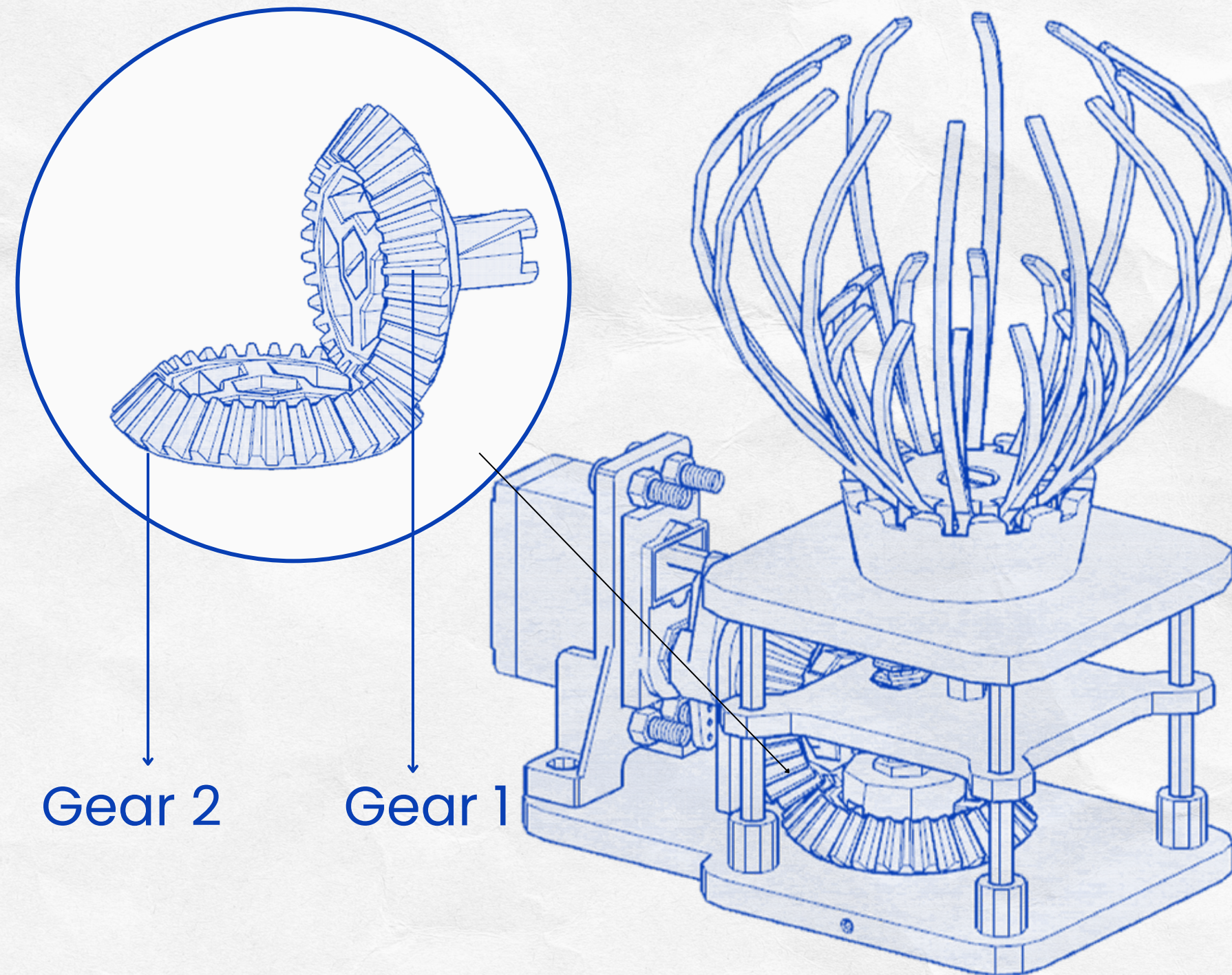
How do the mechanical parts move like flower?

2. How to generate a vertical up-and-down force?



How do the mechanical parts move like flower?

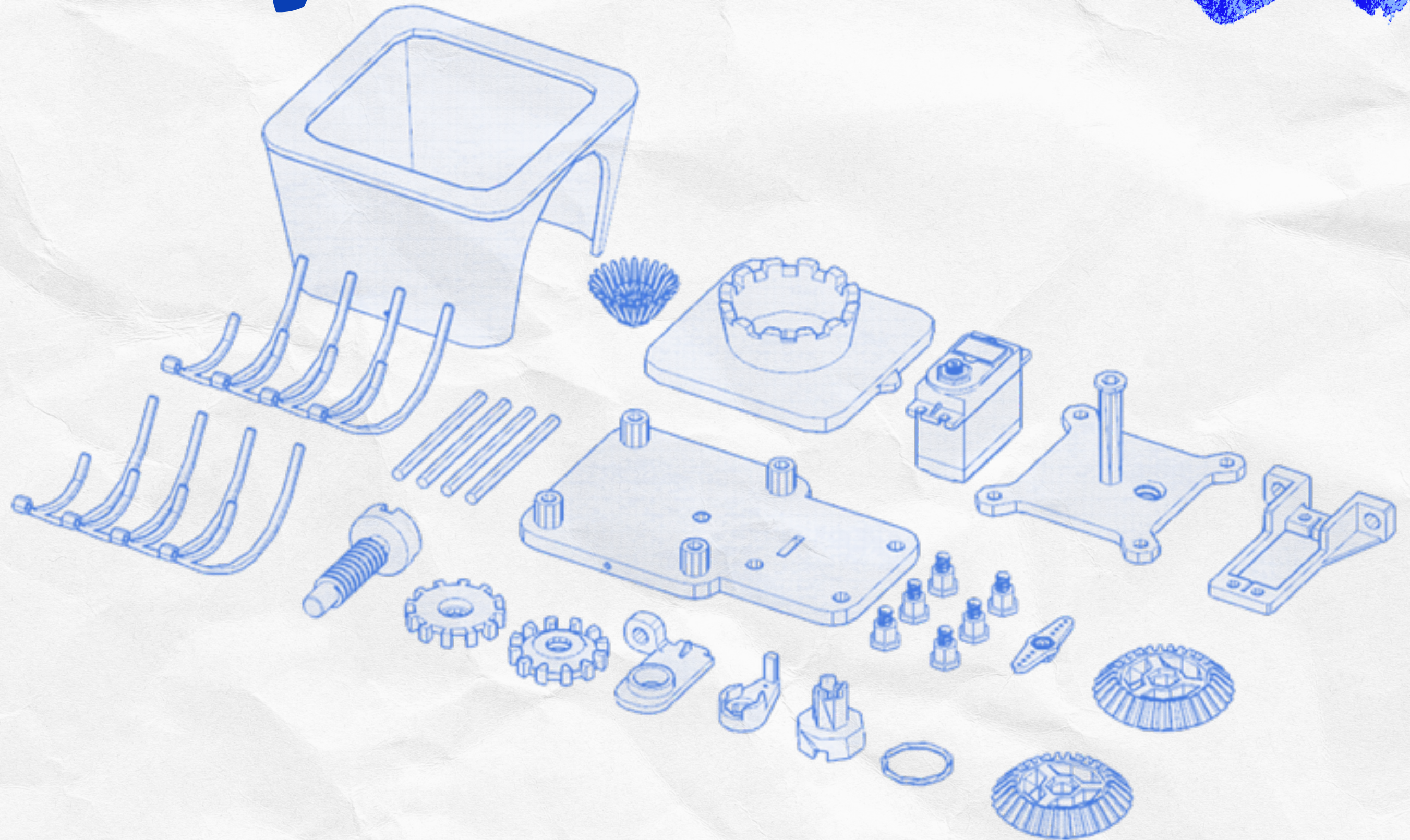
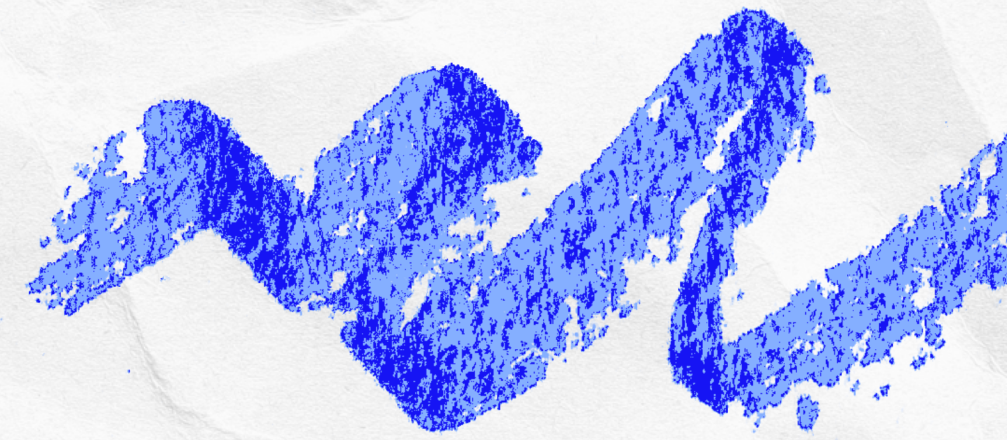
3. Gear transmission



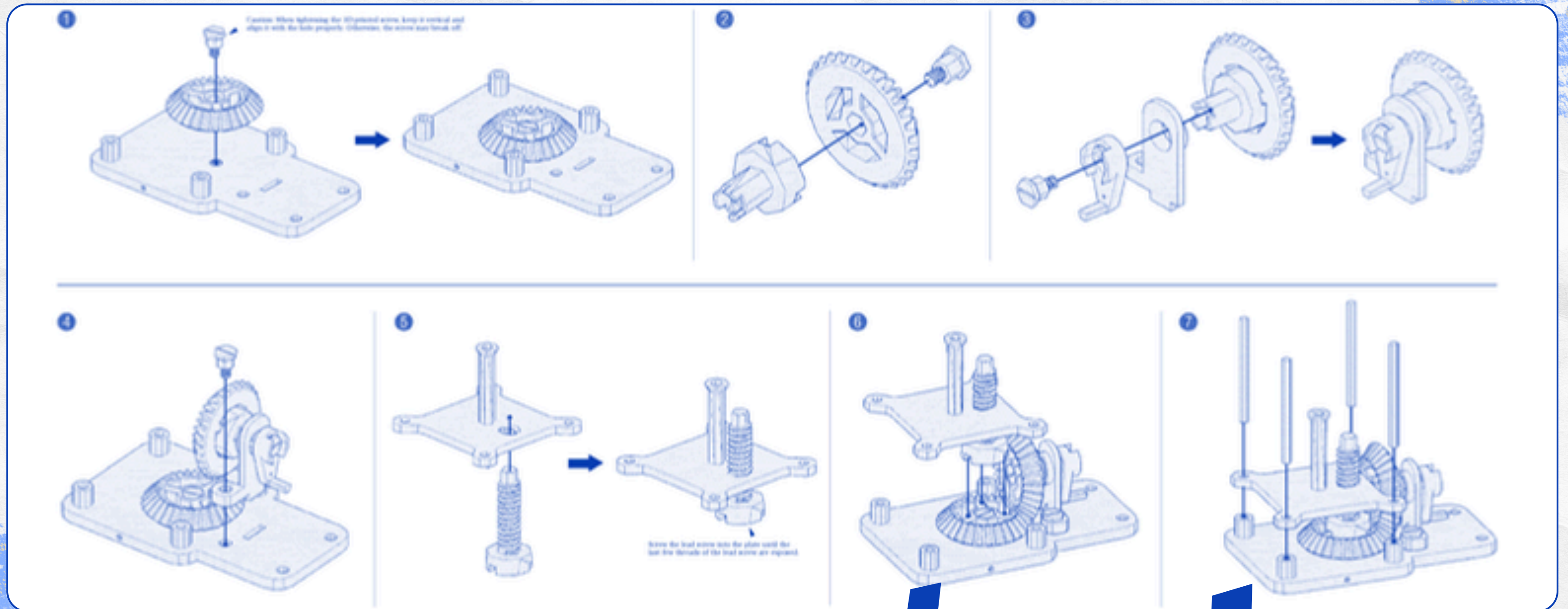
Why?

Change the Direction of Power:
from vertical to horizontal.

3D Printing PARTS



Please check the guidance sheet on your table.

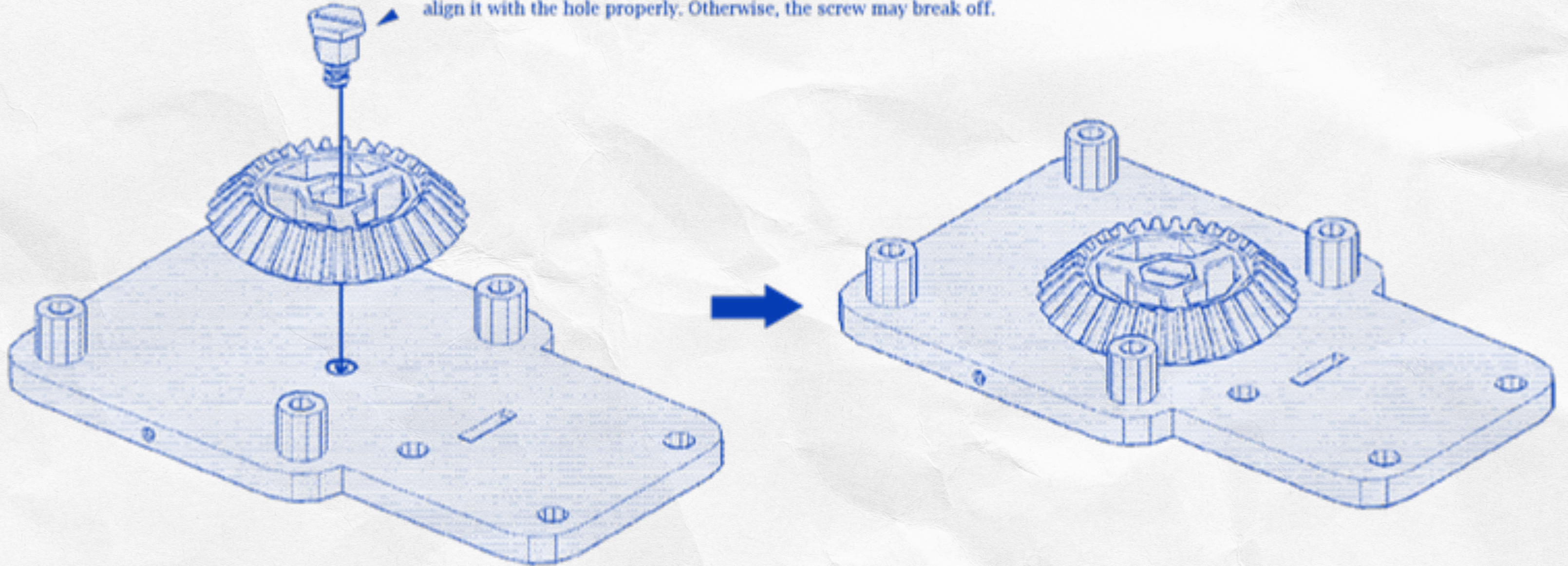


ASSEMBLE it!

Let's ASSEMBle it!

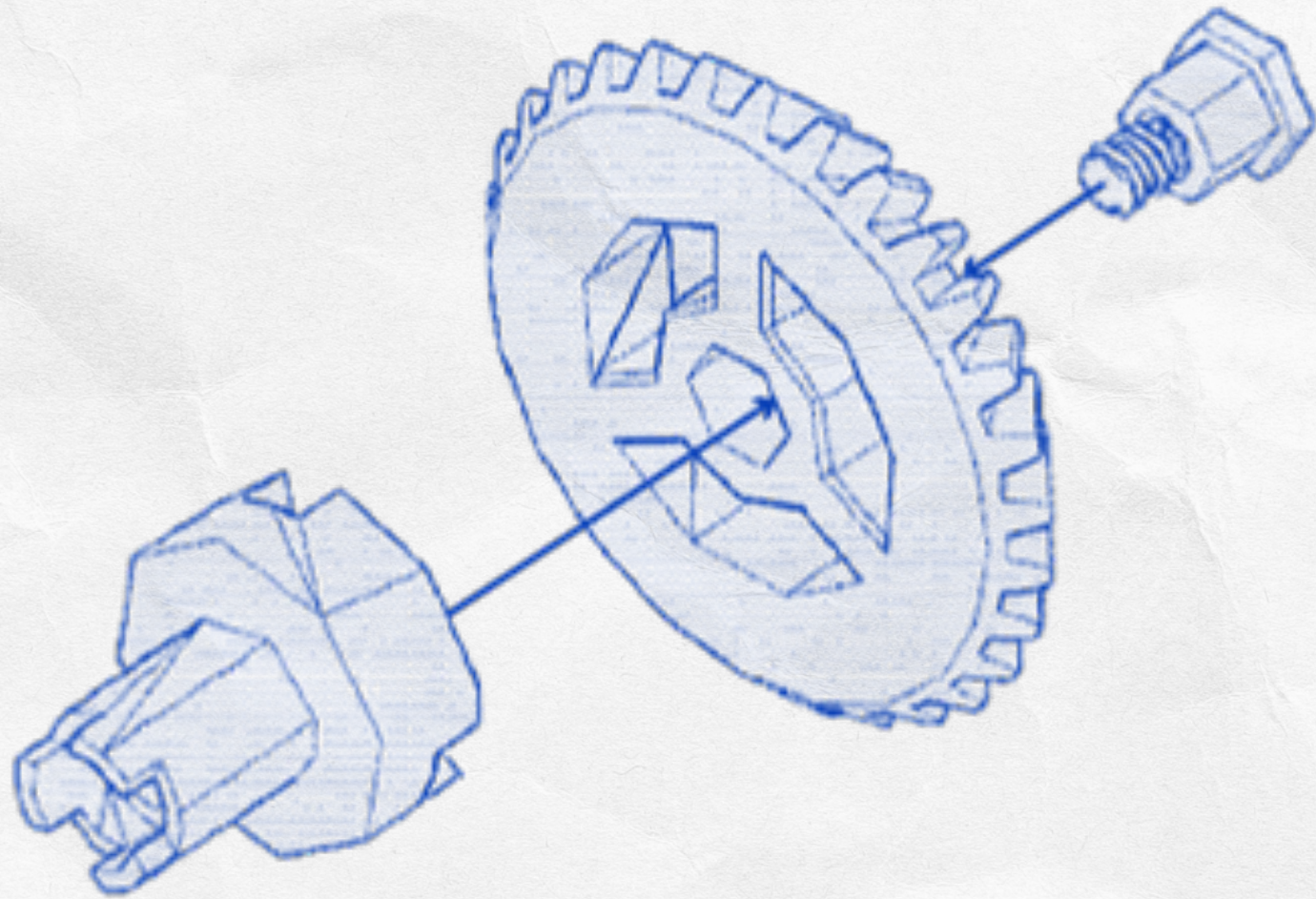
1

Caution: When tightening the 3D-printed screw, keep it vertical and align it with the hole properly. Otherwise, the screw may break off.

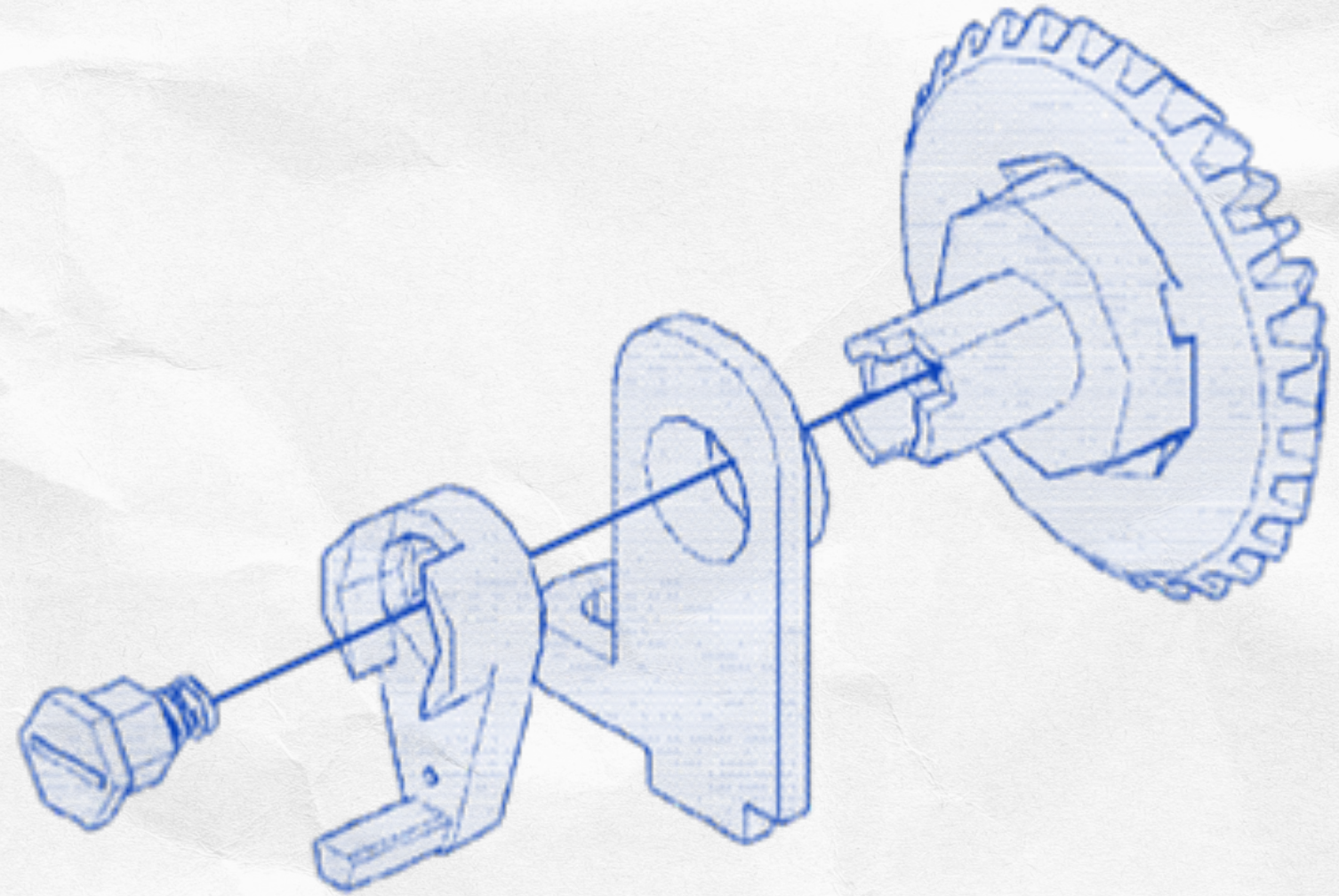


Let's ASSEMBle it!

2

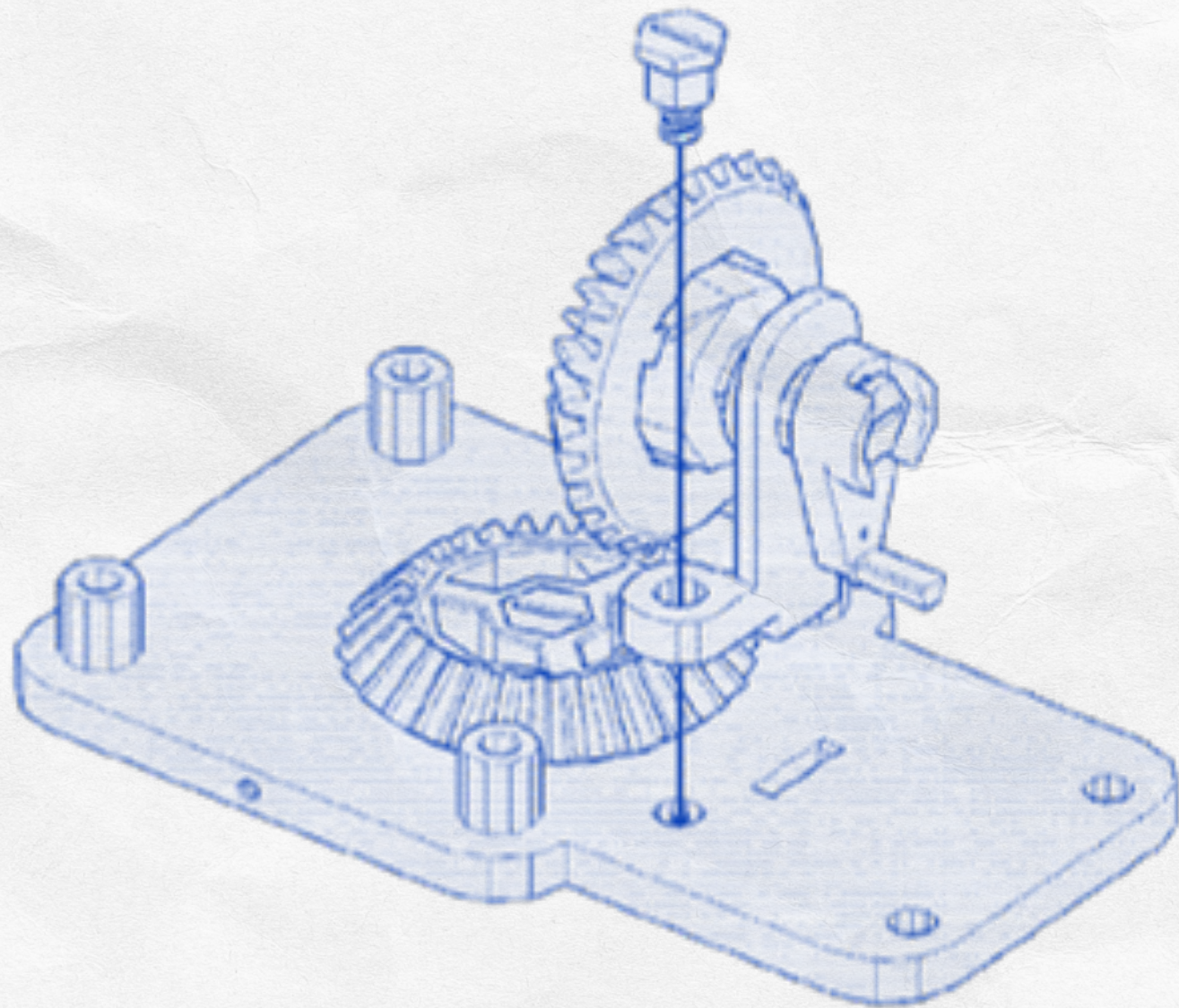


3

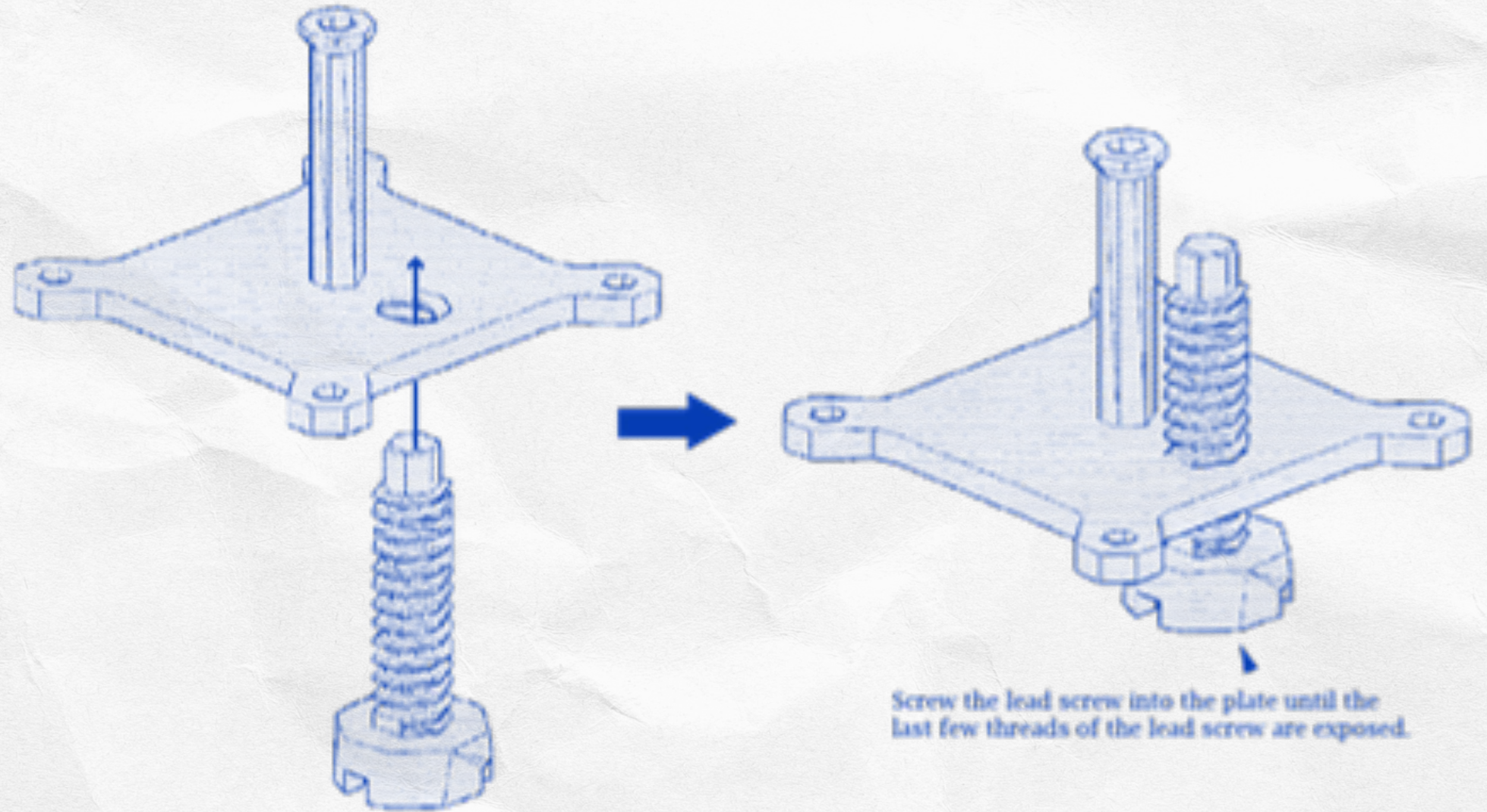


Let's ASSEMBle it!

4



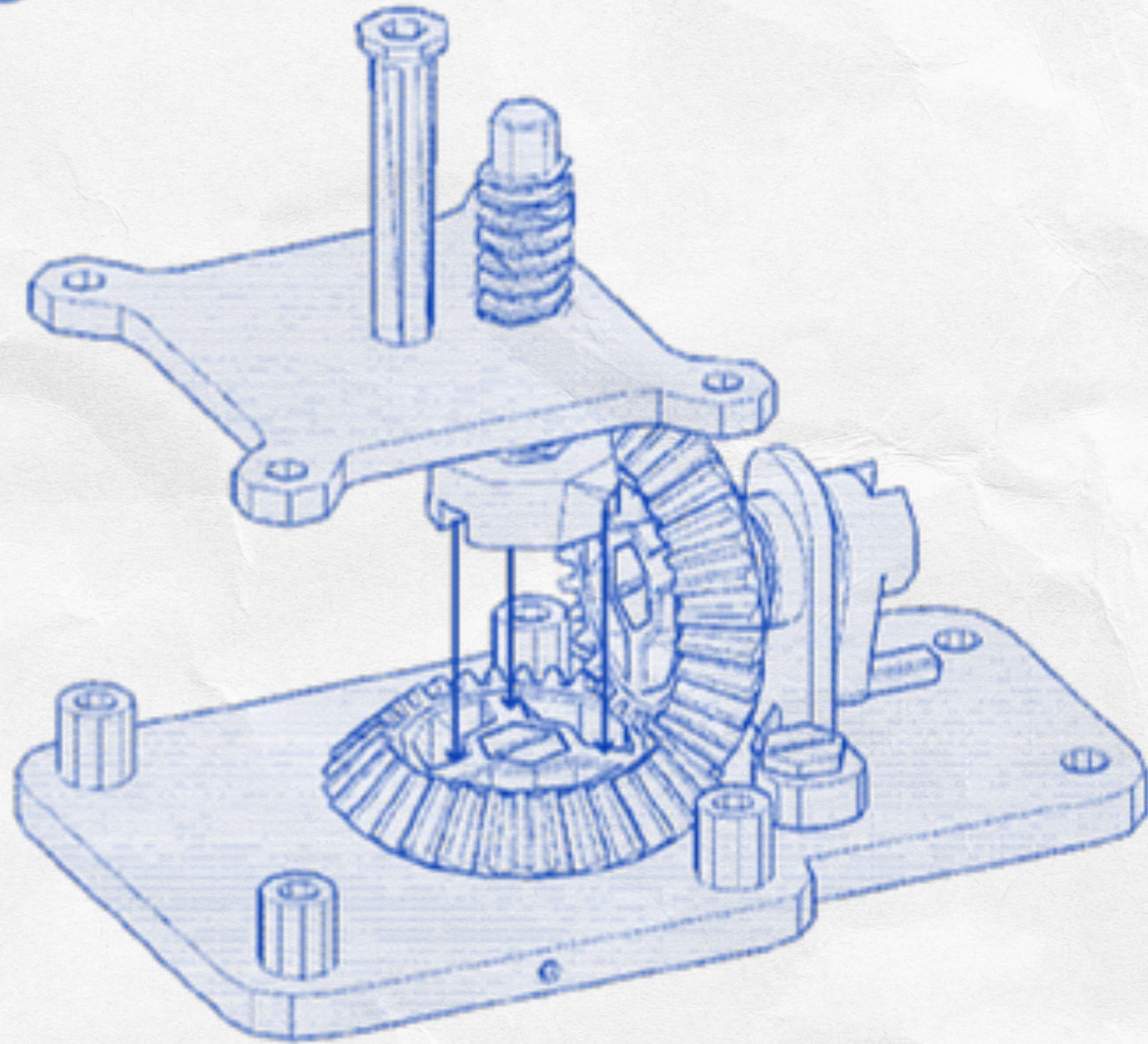
5



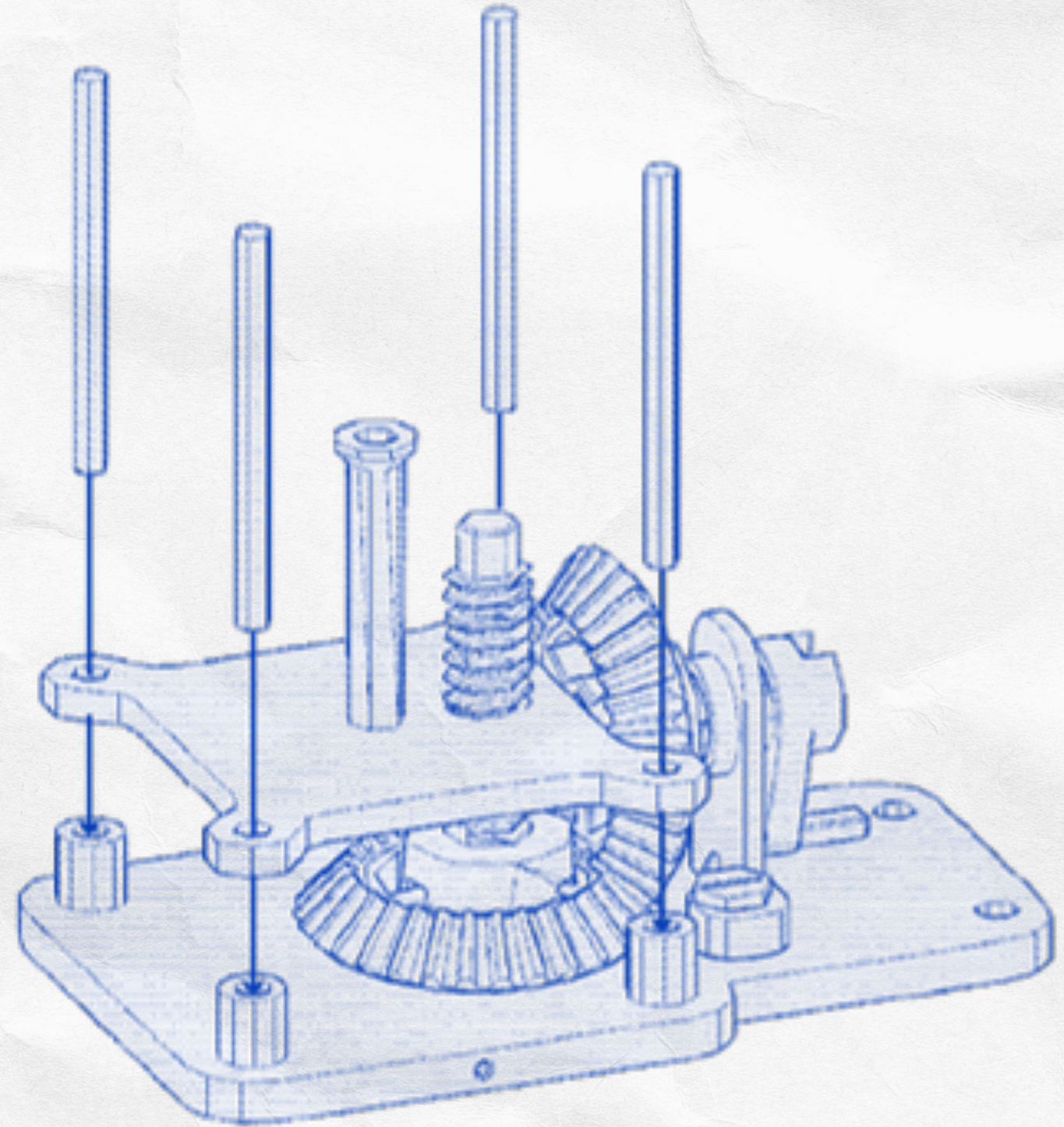
Screw the lead screw into the plate until the last few threads of the lead screw are exposed.

Let's ASSEMBle it!

6

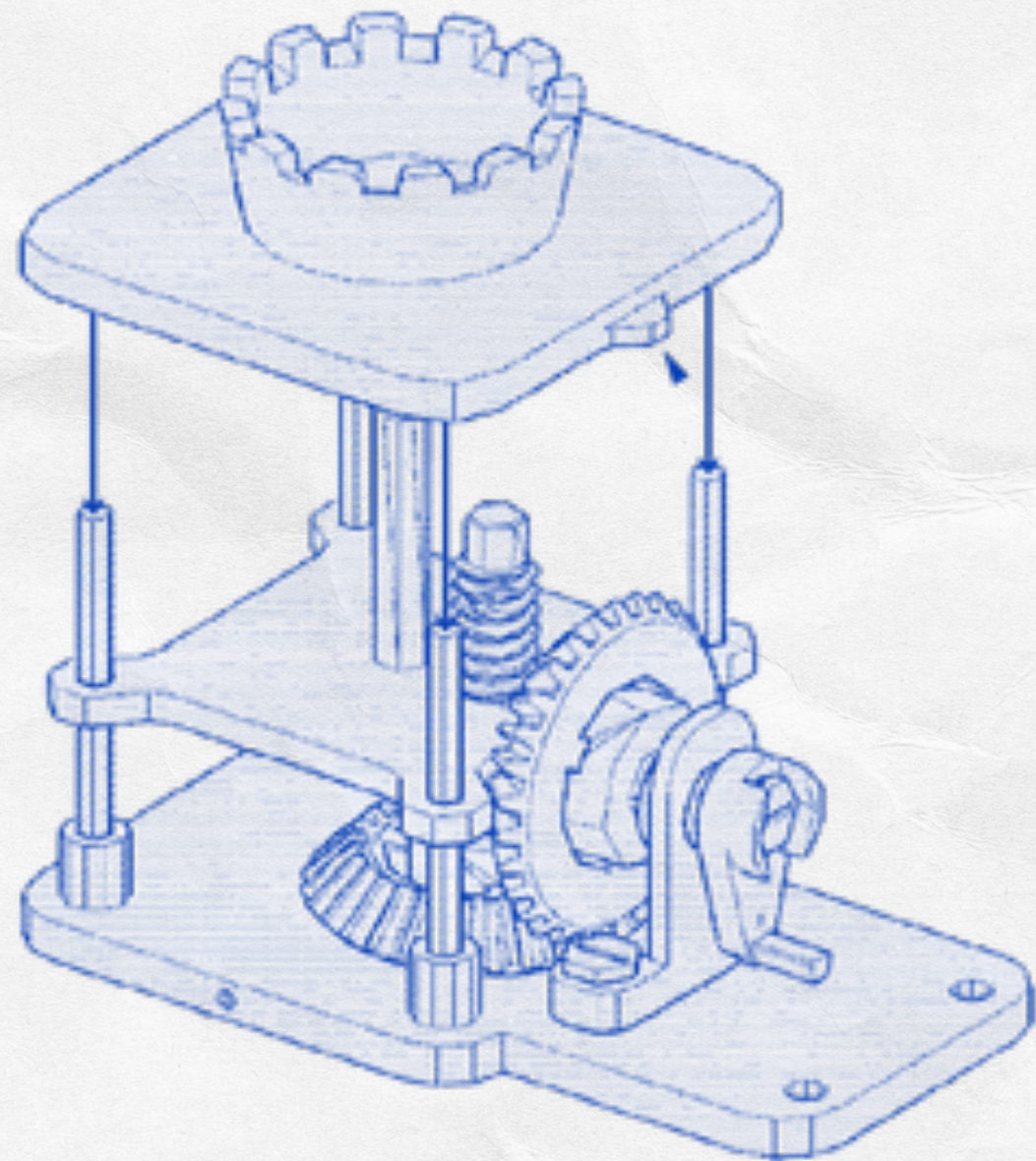


7

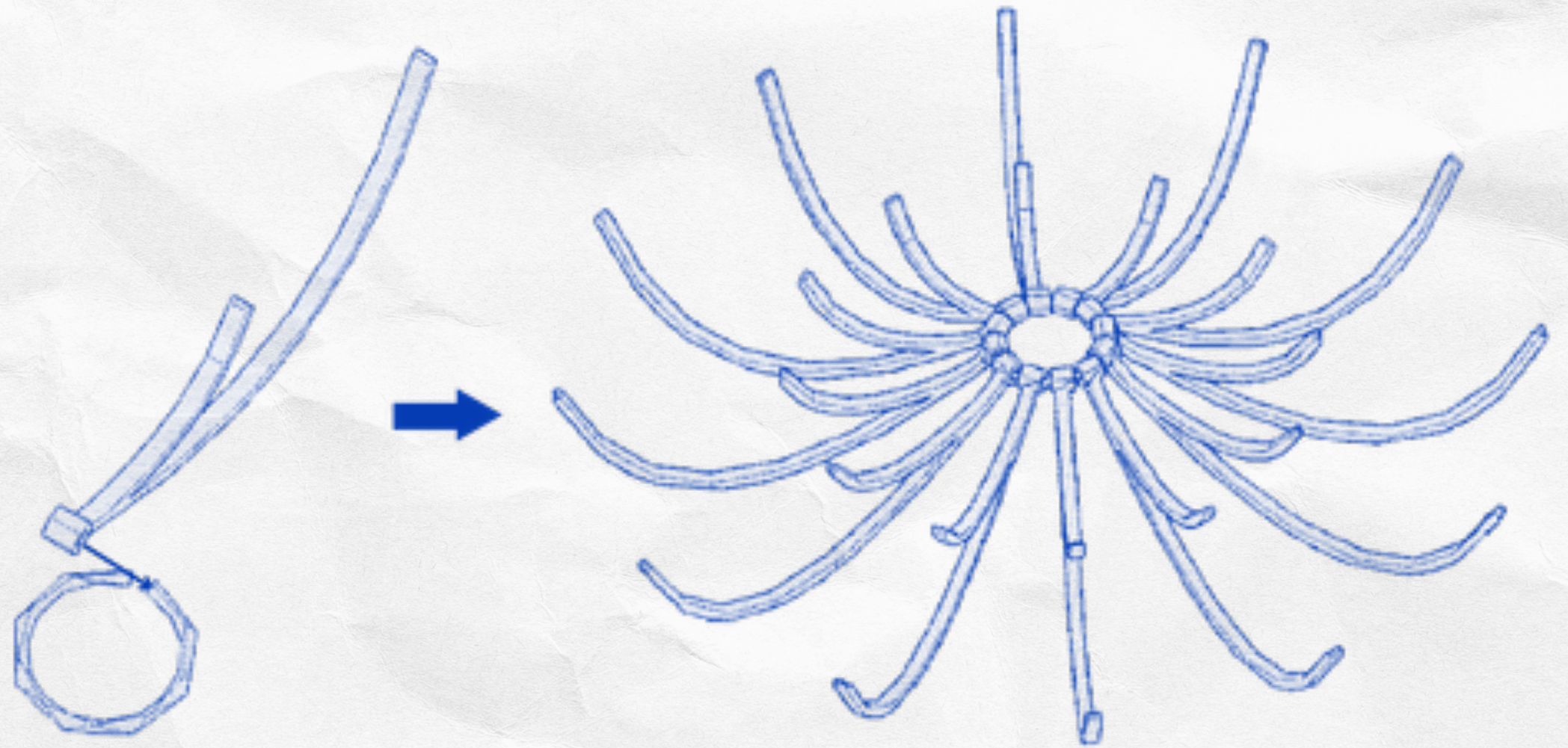


Let's ASSEMBle it!

8



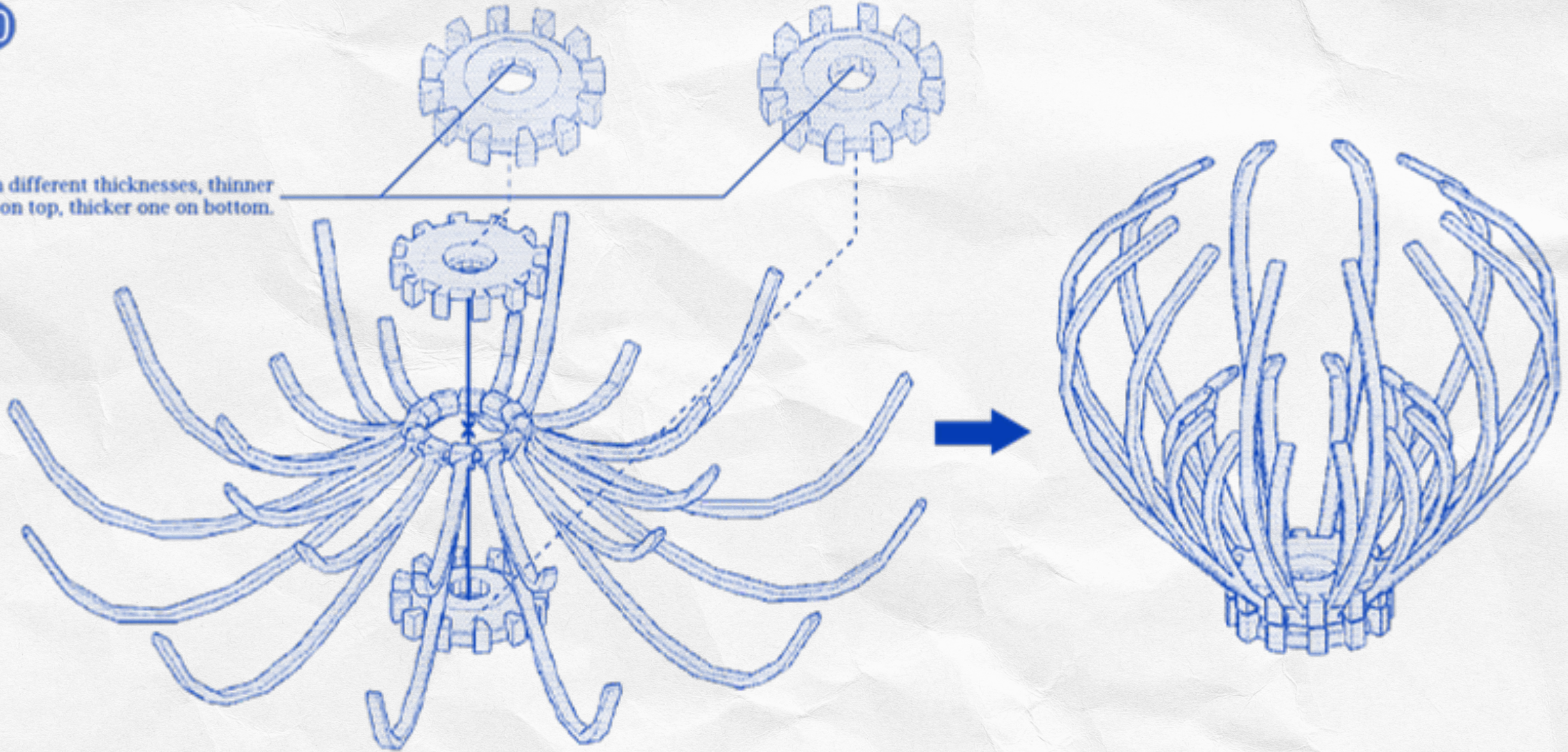
9



Let's ASSEMBle it!

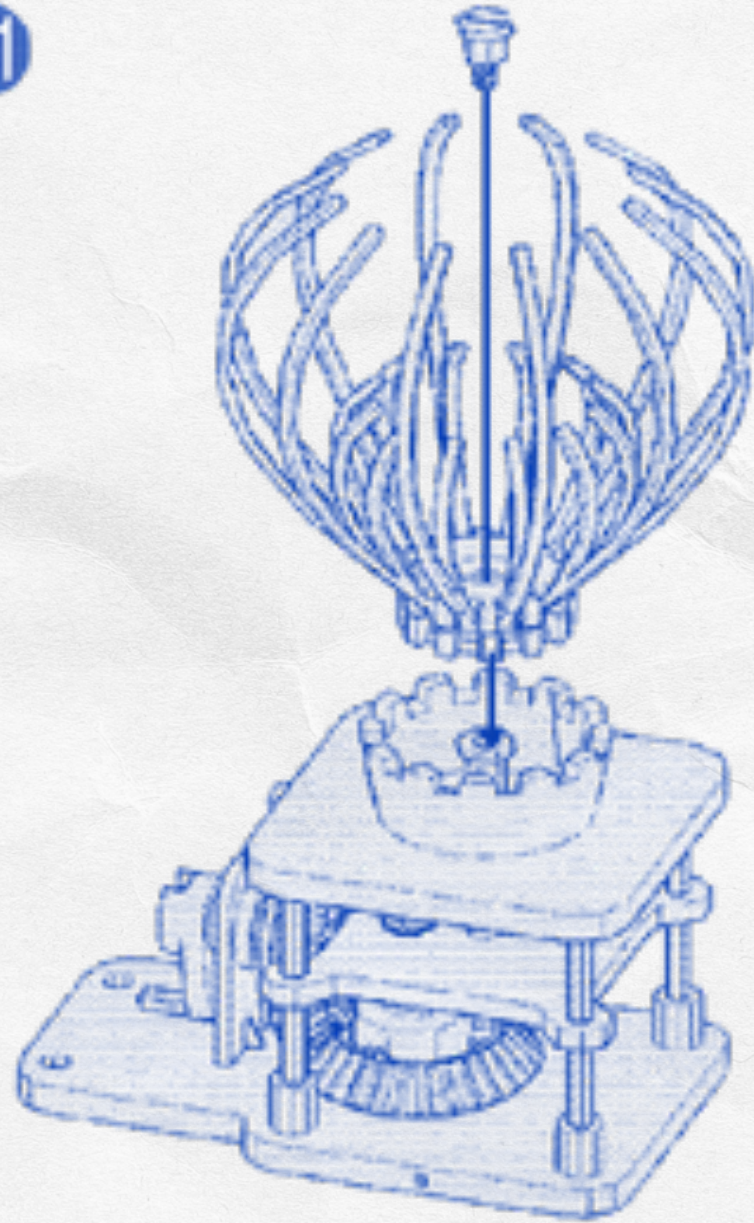
10

With different thicknesses, thinner
one on top, thicker one on bottom.



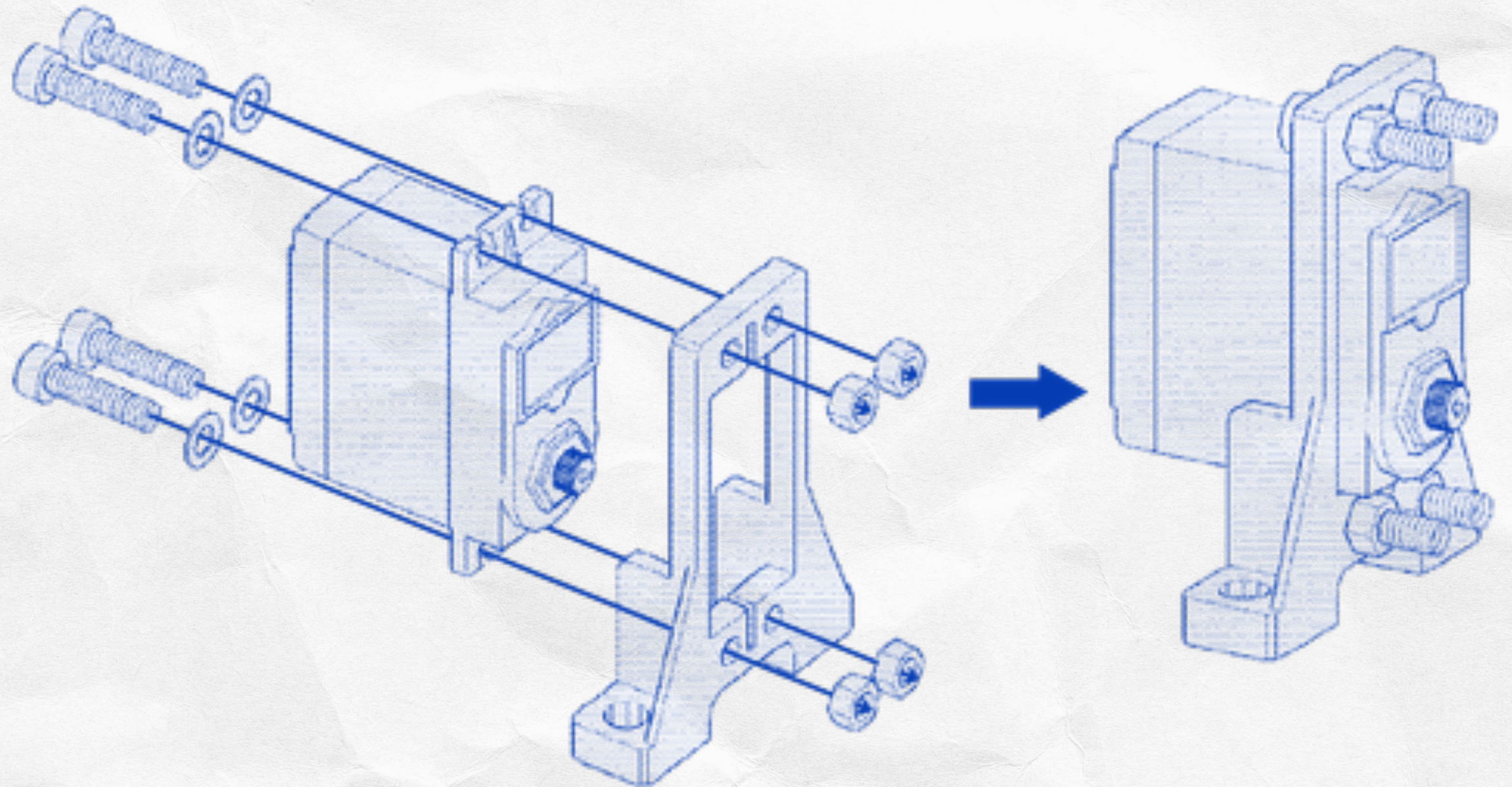
Let's ASSEMBLE it!

11



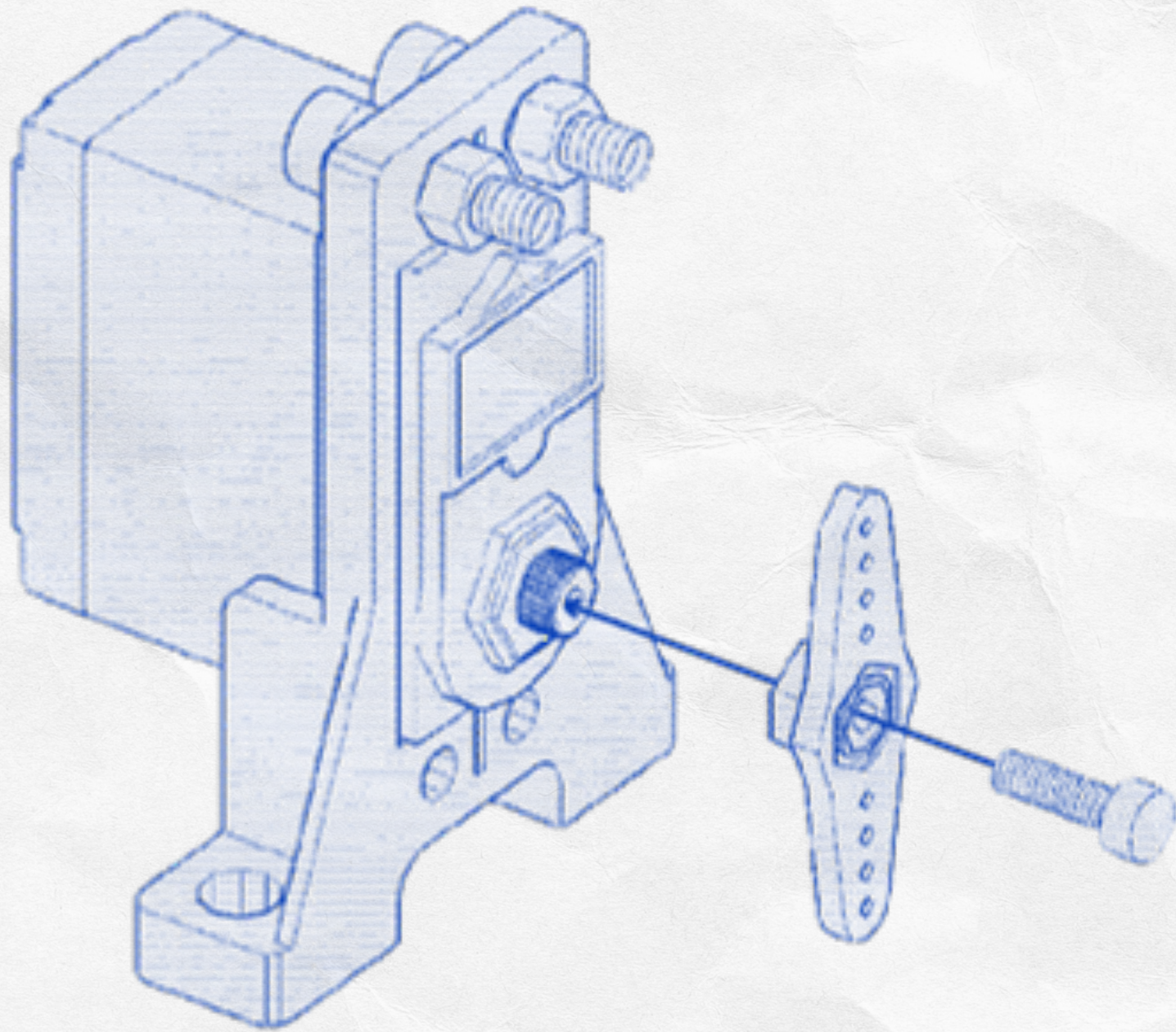
Now you can manually rotate to make the petals move.

12

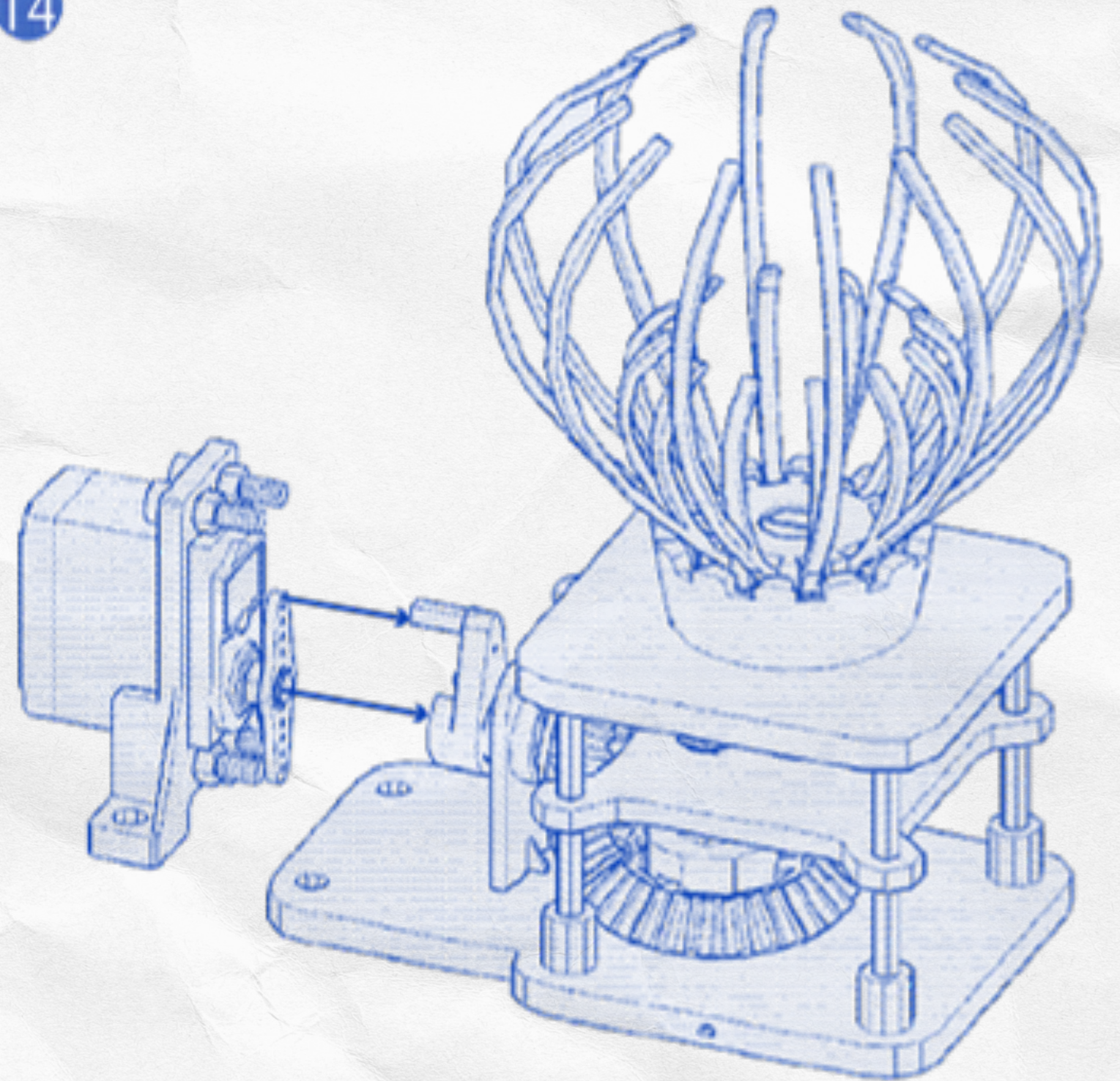


Let's ASSEMBle it!

13

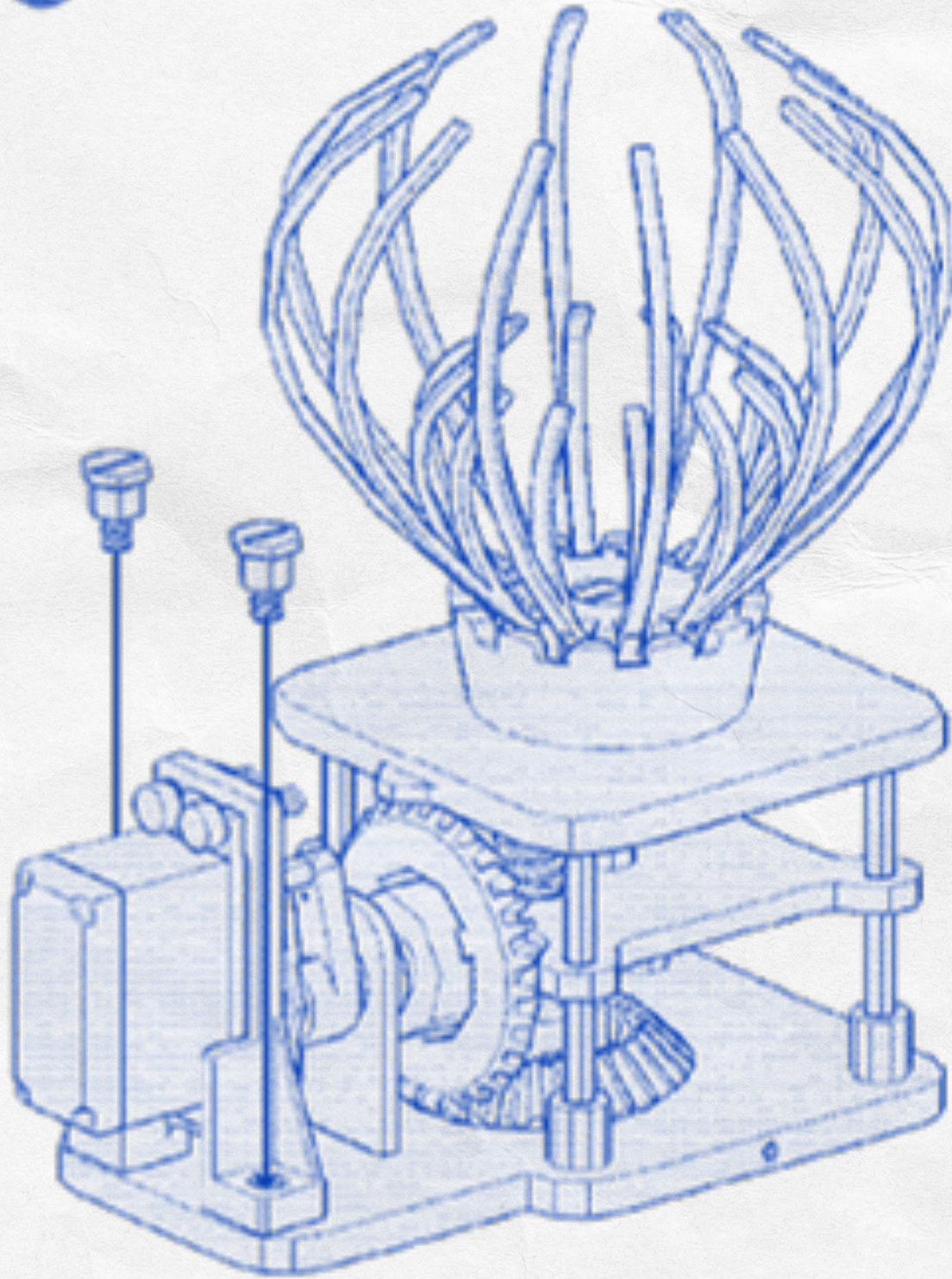


14

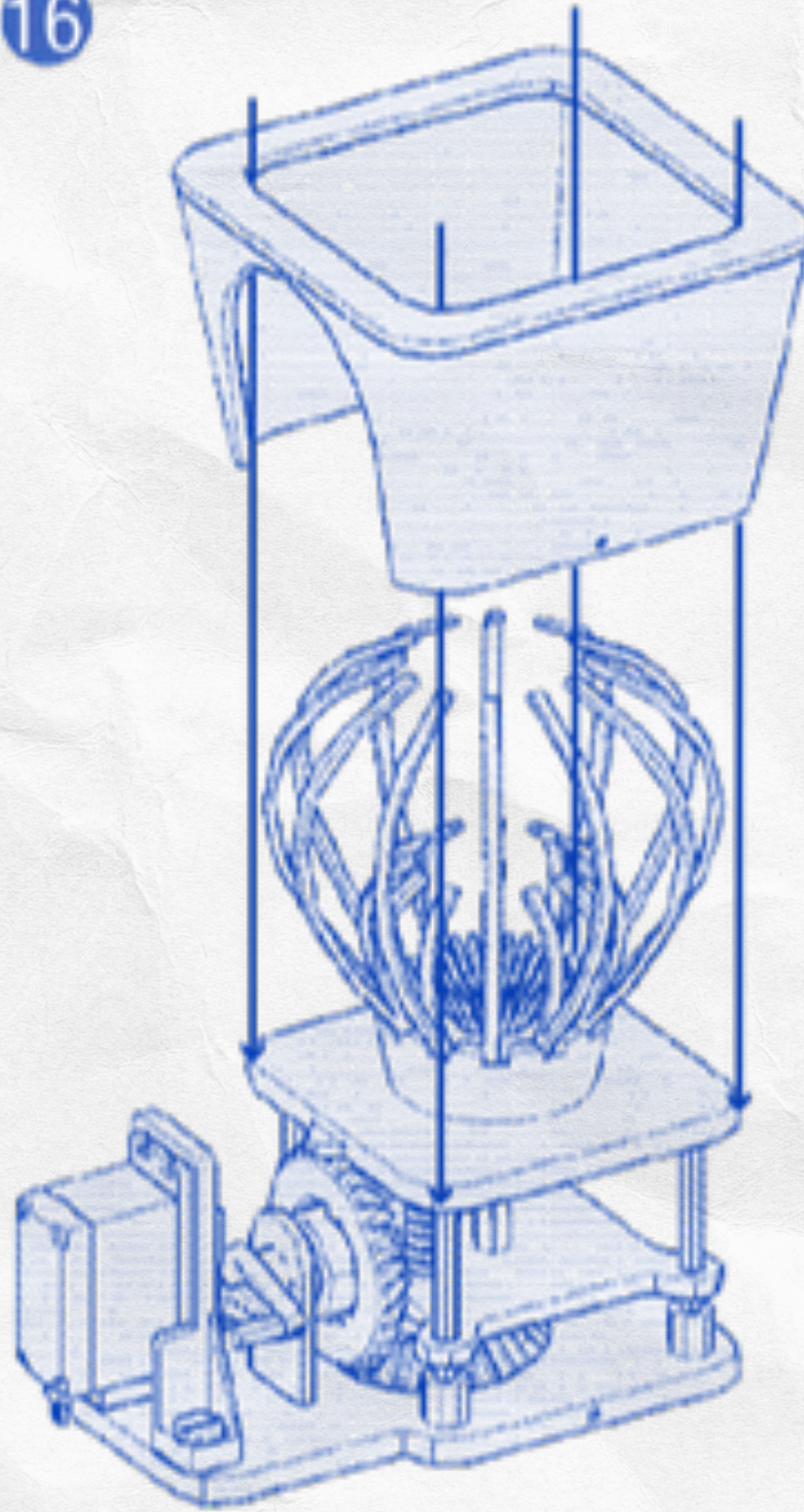


Let's ASSEMBLE it!

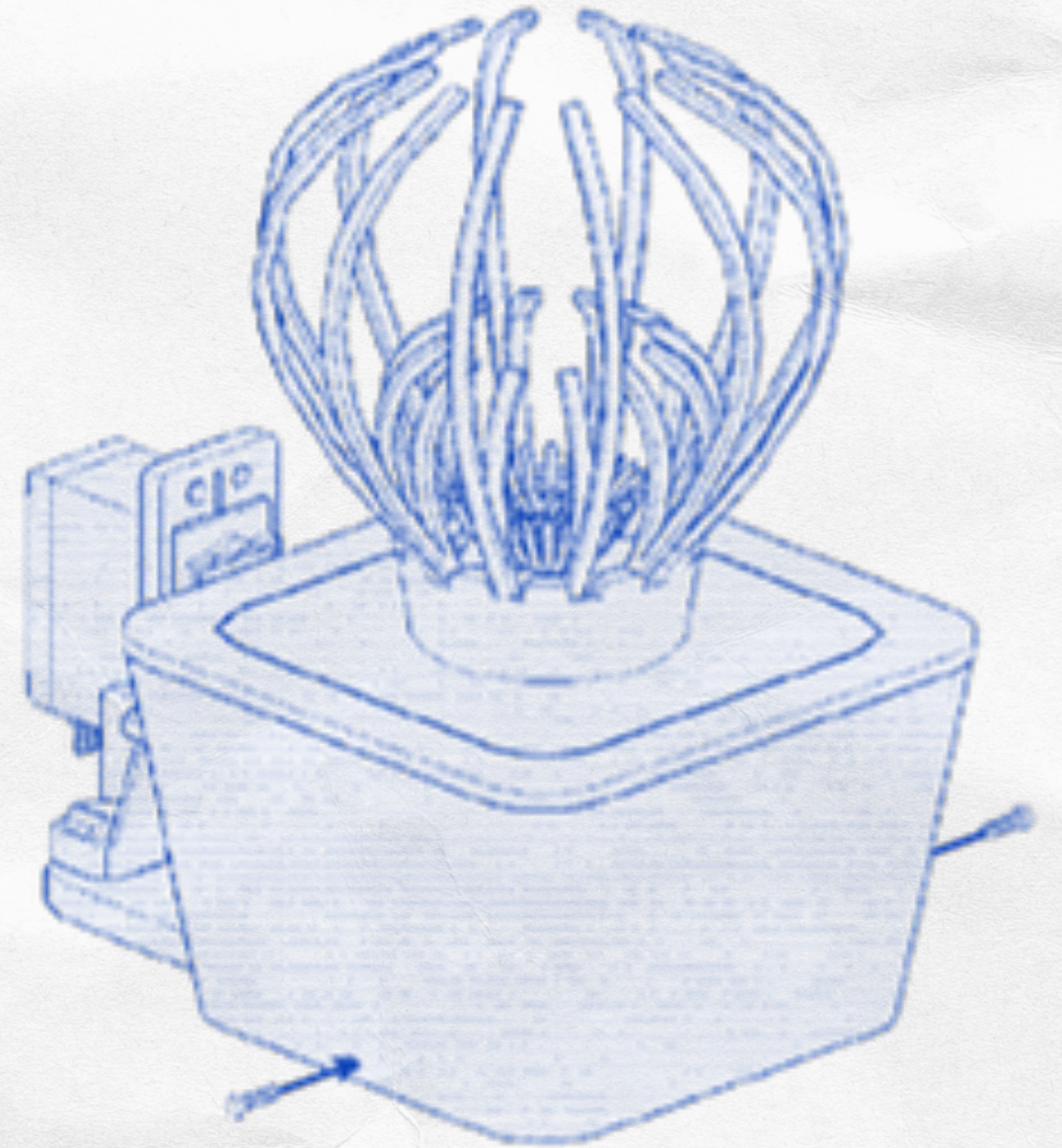
15



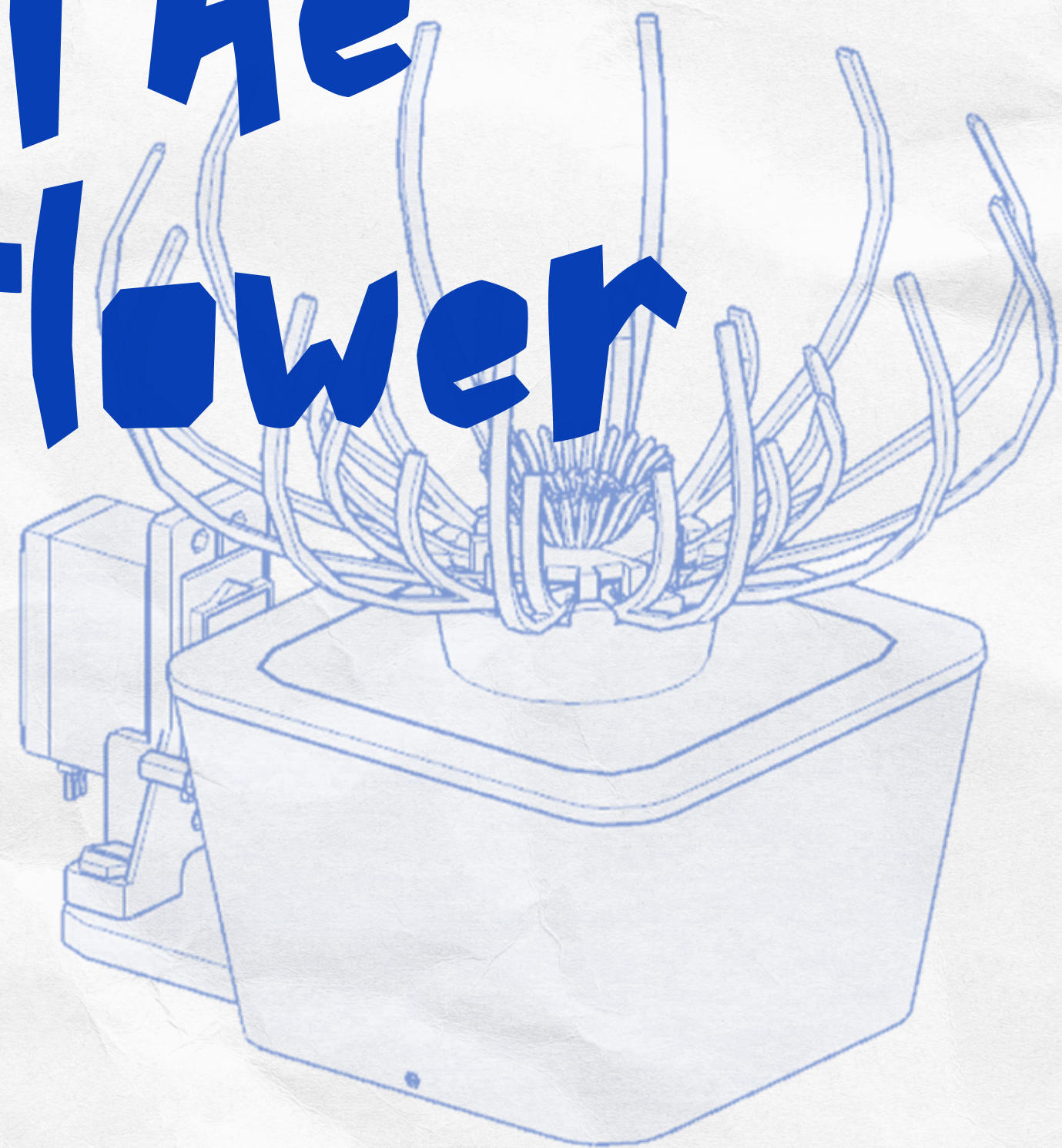
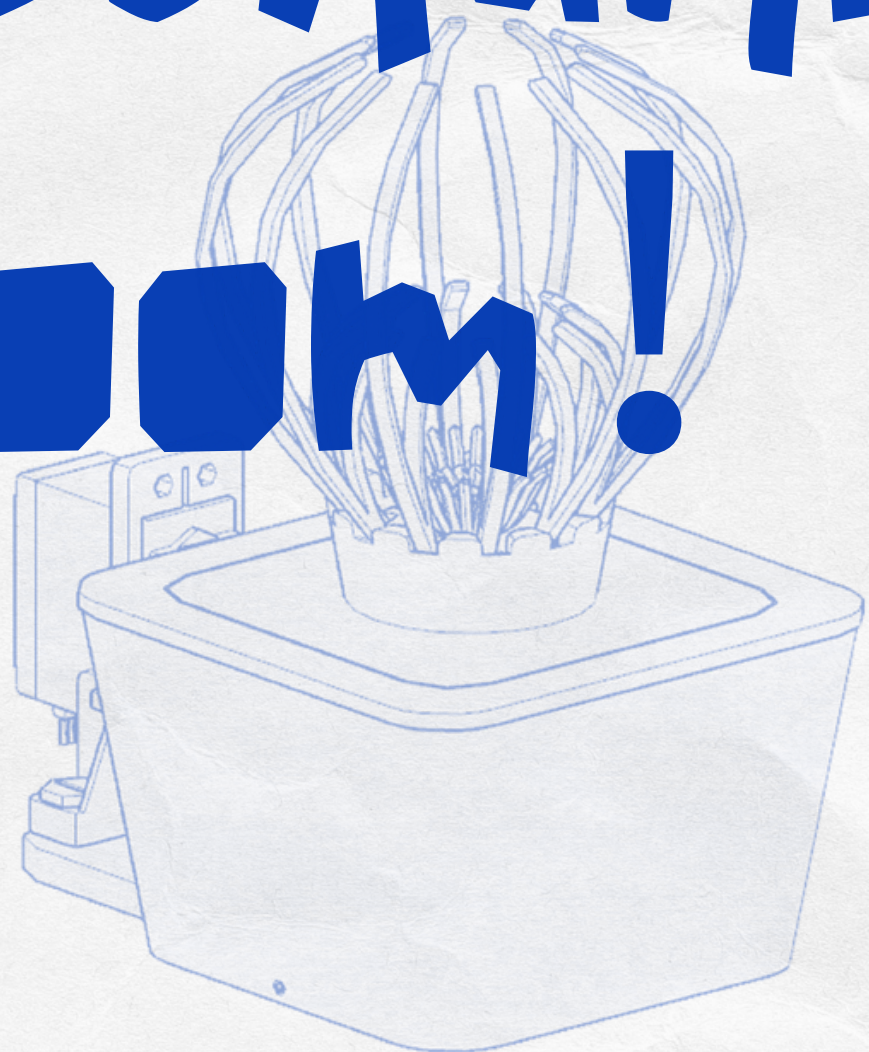
16



17



Let's Make the
Mechanical Flower
Bloom!



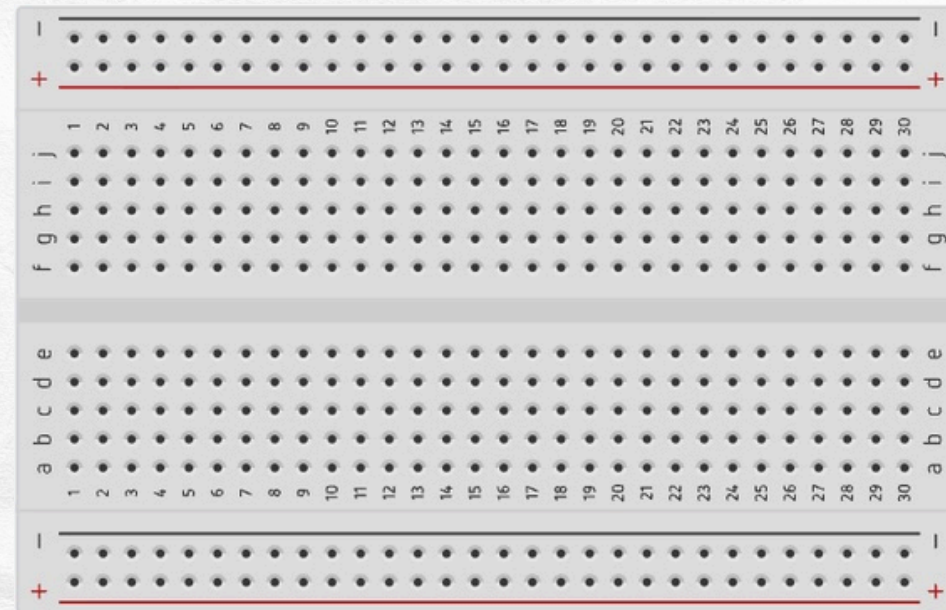


Let's check together!

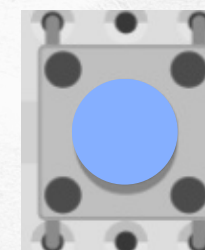
Checklist



Arduino Nano



Bread Board



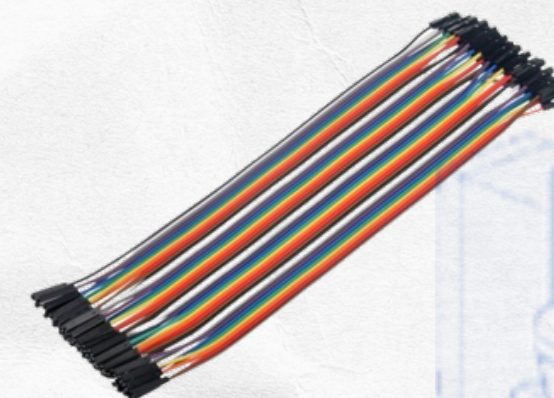
Button x 2



Servo MG996R



Battery

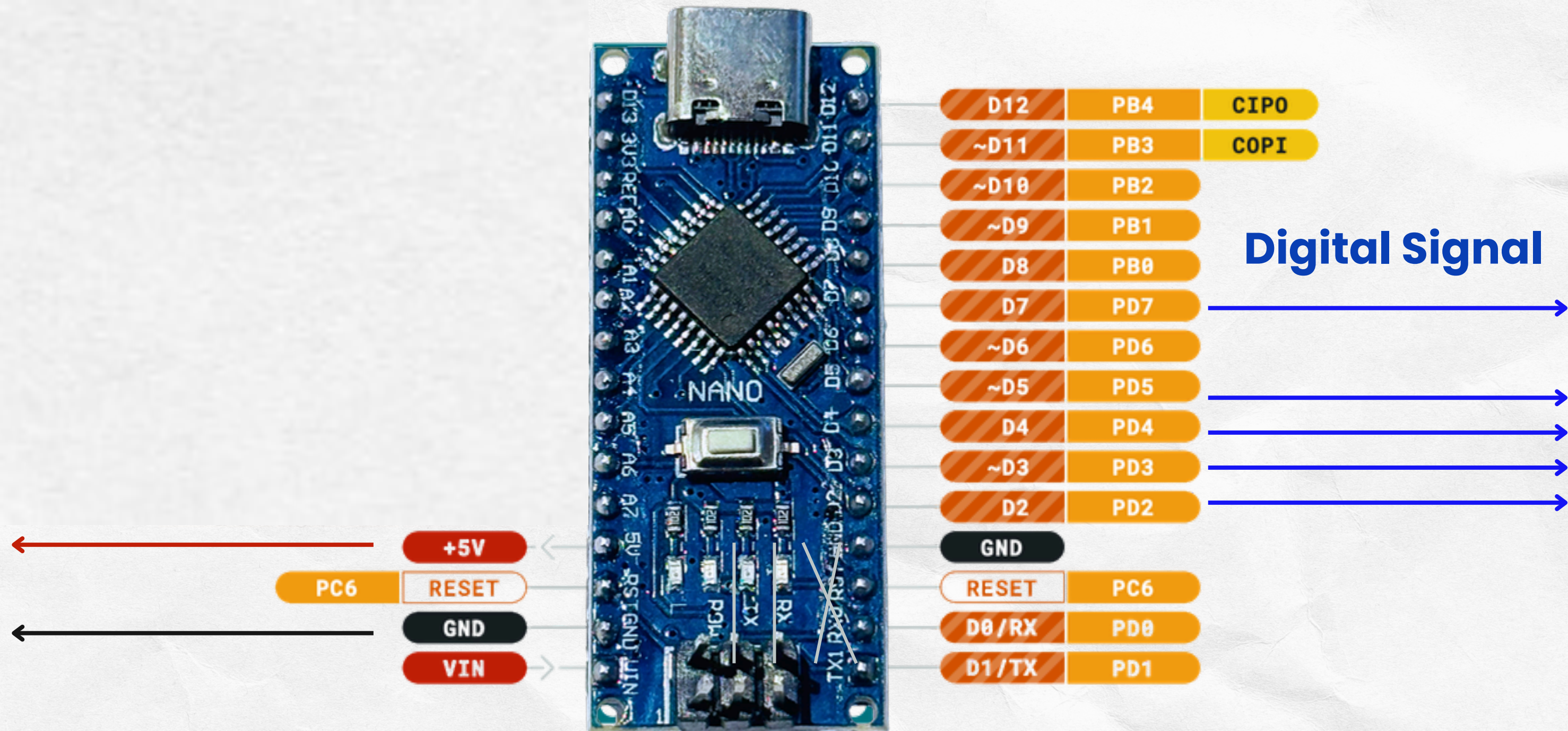


Dupont Lines/Skip Lines



Arduino Nano Pinout Diagram

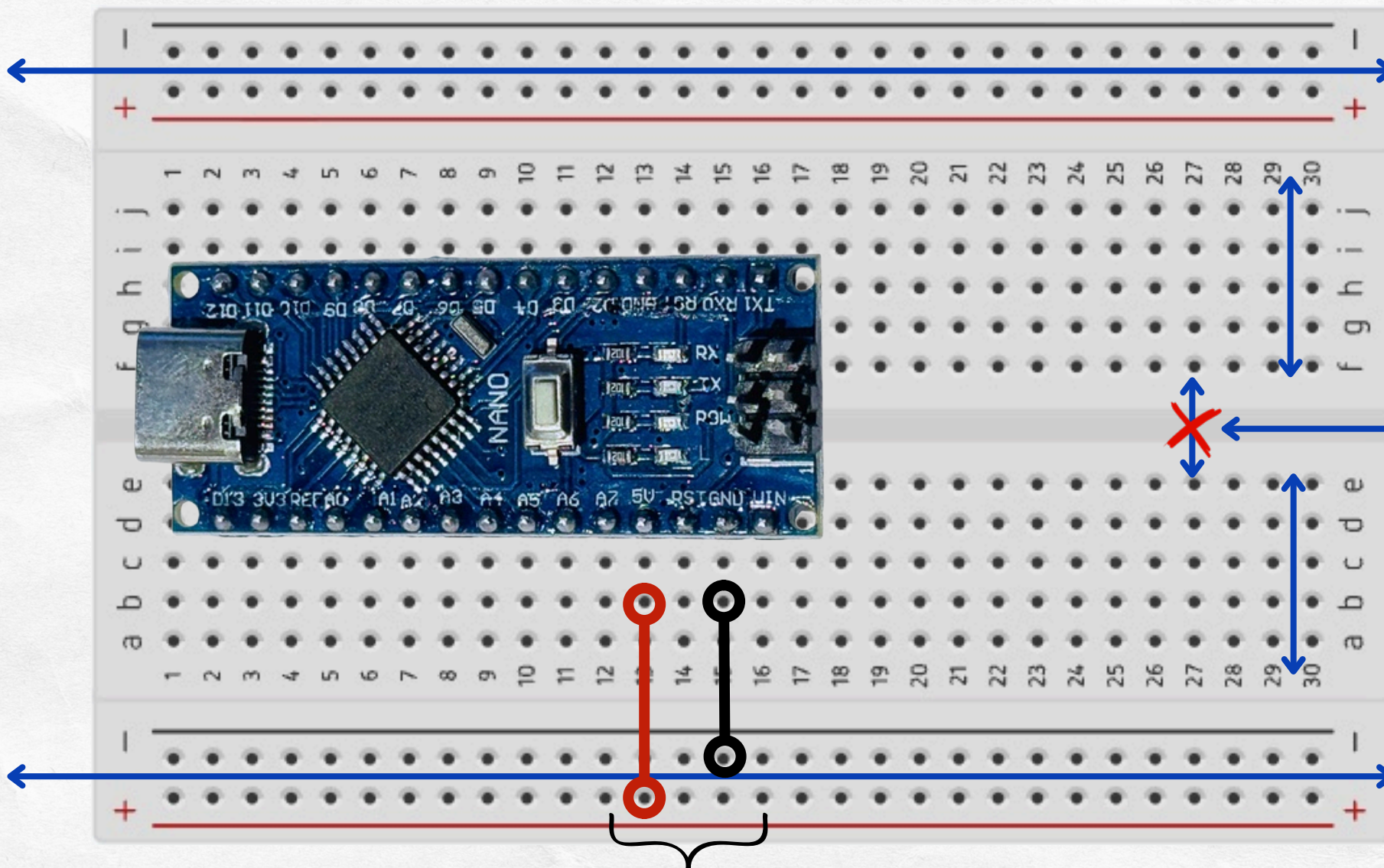
"Brain"





Bread Board ——LEGO baseplate of electronics

“plug-and-play”



Power Rails (Horizontally Connected),
1 through 30 fully linked, no connection between each other rail

Nodes, Each column is vertically connected,
no connection between each strip, between e and f

Center Gap Channel

Power Rails (Horizontally Connected),
1 through 30 fully linked, no connection between the two rails

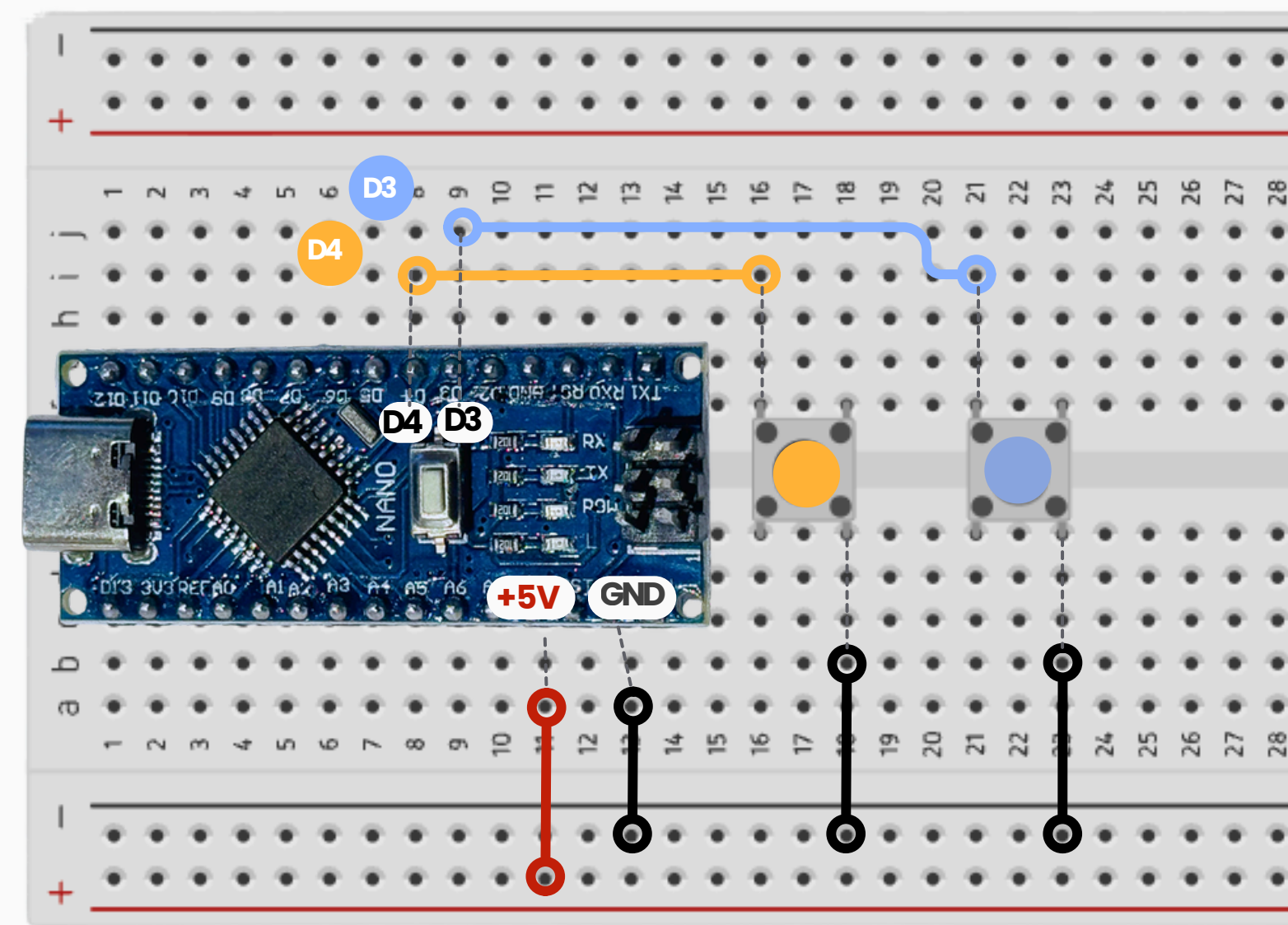
Export **+5V** and **GND** on Bread Board



BUTTON

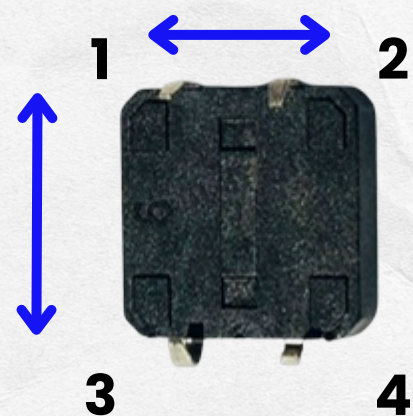
4 pins button? No, just for stability.

Demo



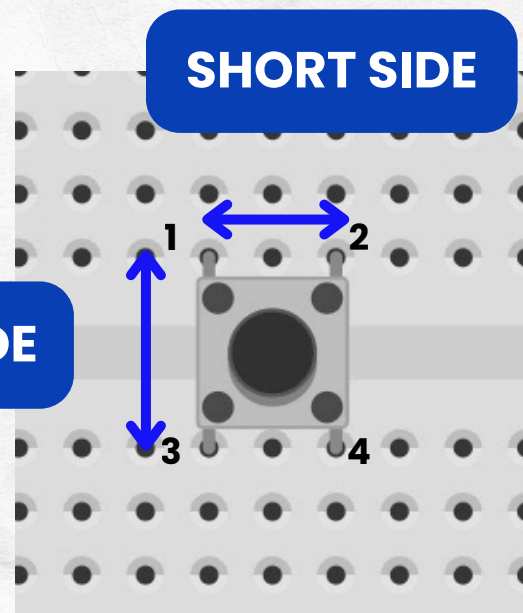
SHORT SIDE

LONG SIDE



SHORT SIDE

LONG SIDE



$V1=V2; V3=V4$

pin: 1/2



3/4

Not Pressed

1/2



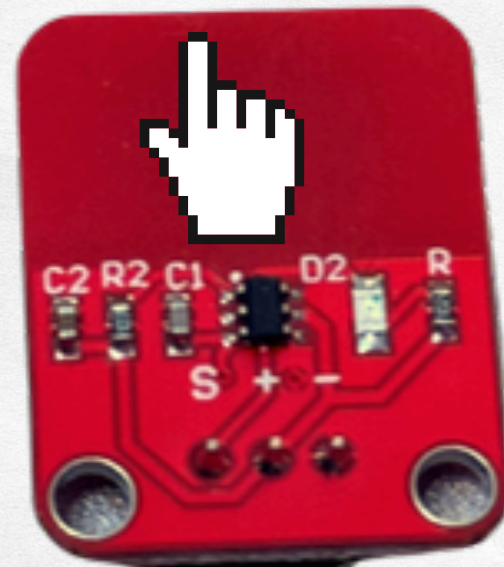
3/4

Pressed



INTERACTIVE TOUCH SENSOR

"One of the plates of Capacitor"

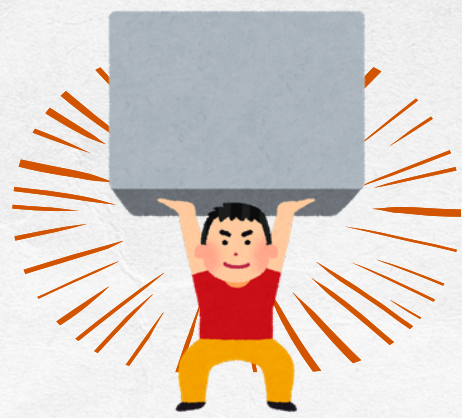


$$(C = \frac{\epsilon S}{4\pi k d})$$

S to D5

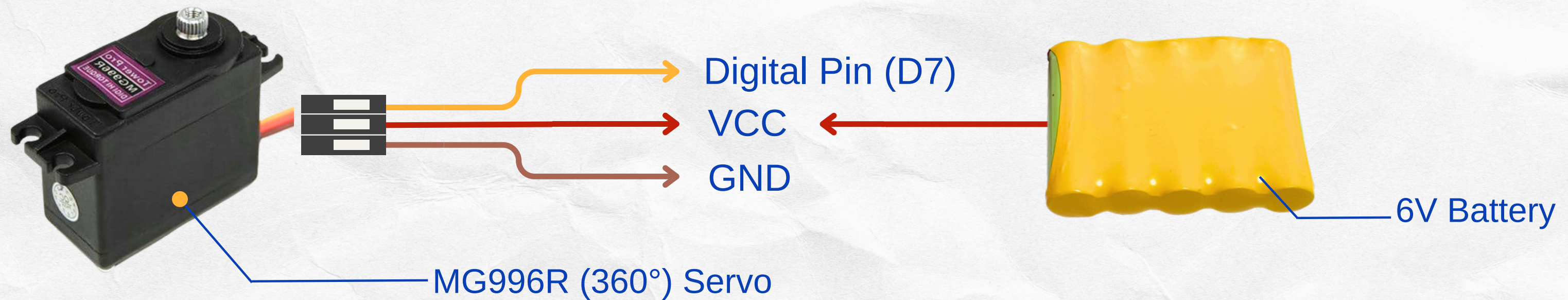
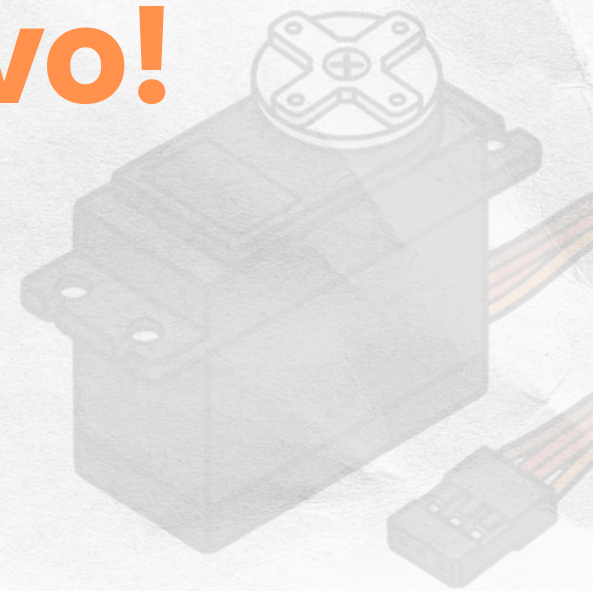
+ 5V

GND



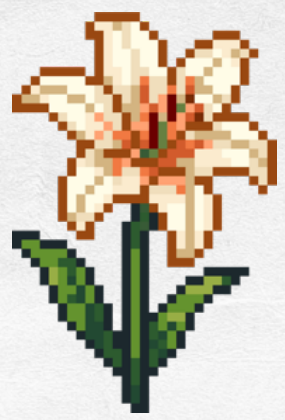
Let's make the flower move with the Servo!

MG996R, Metal Gear(MG)

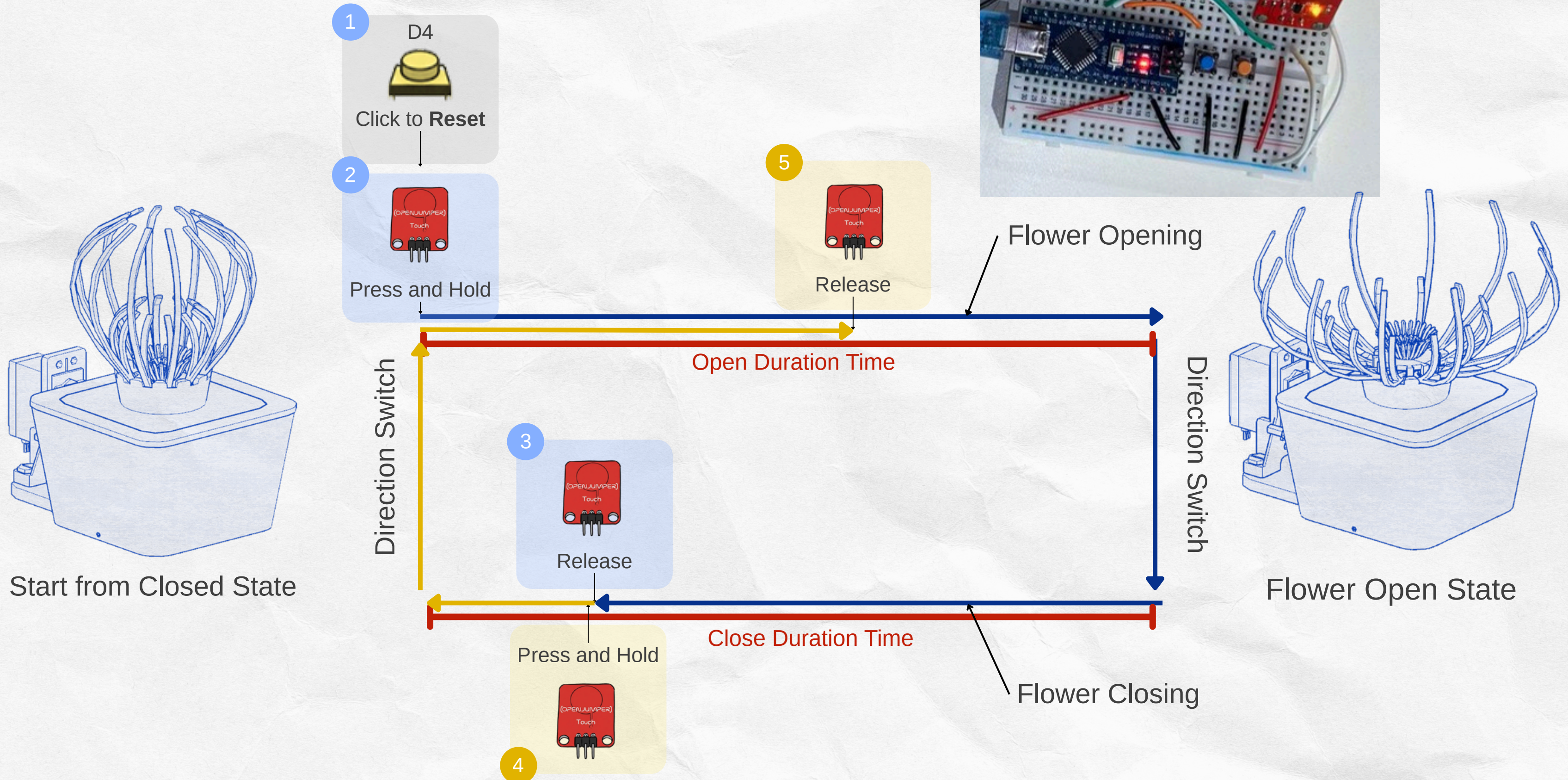
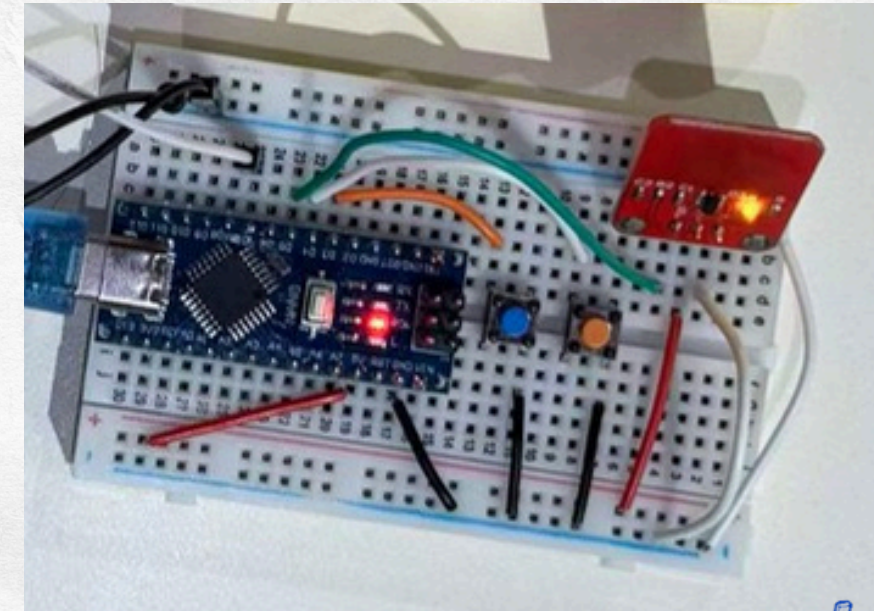


Yellow	D7	Servo Motor
Red	Battery (+) Red	
Black	GND	

✗ Do not connect the servo with 5V pin on Nano

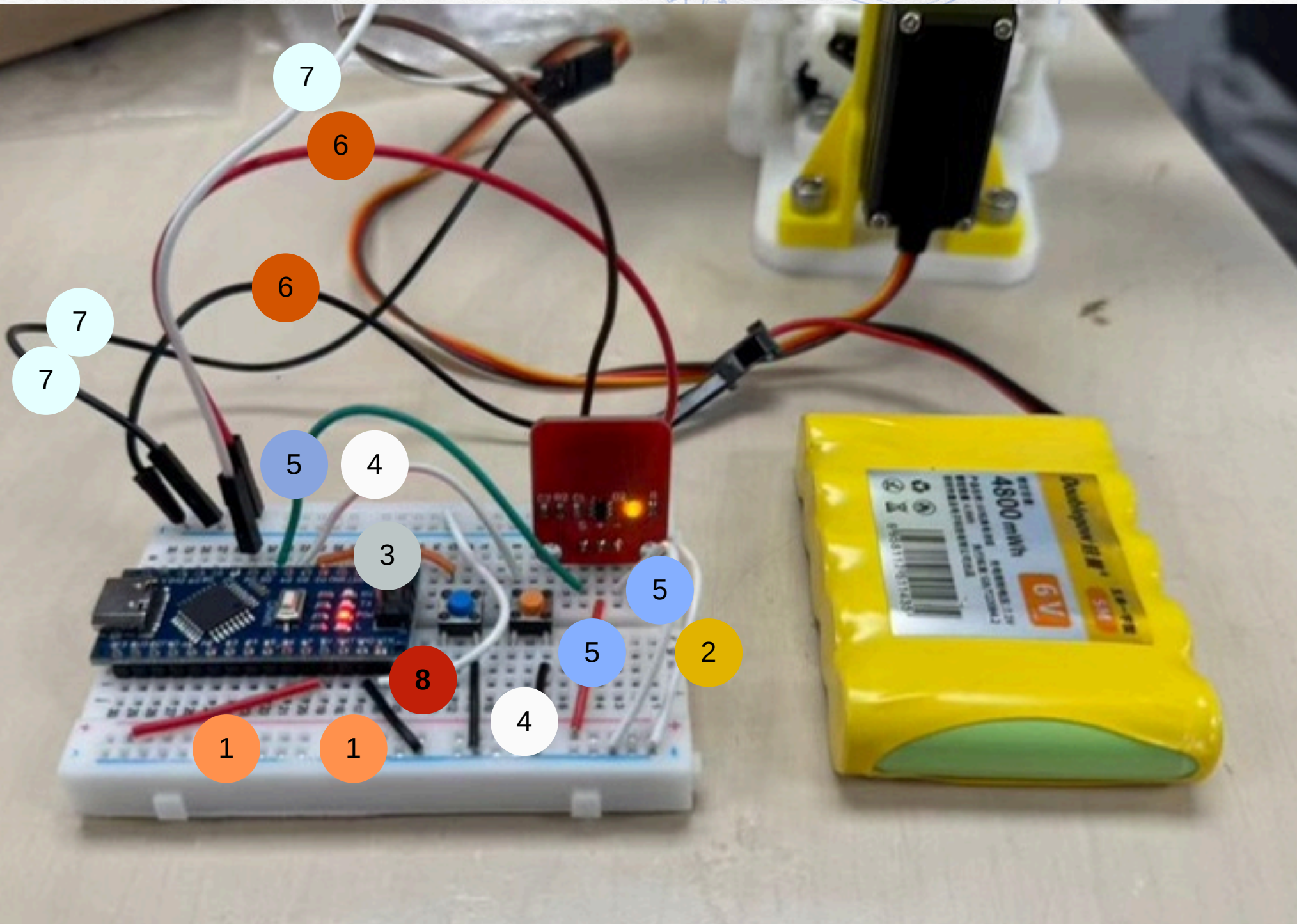


Let's Bloom Flower with Interaction!

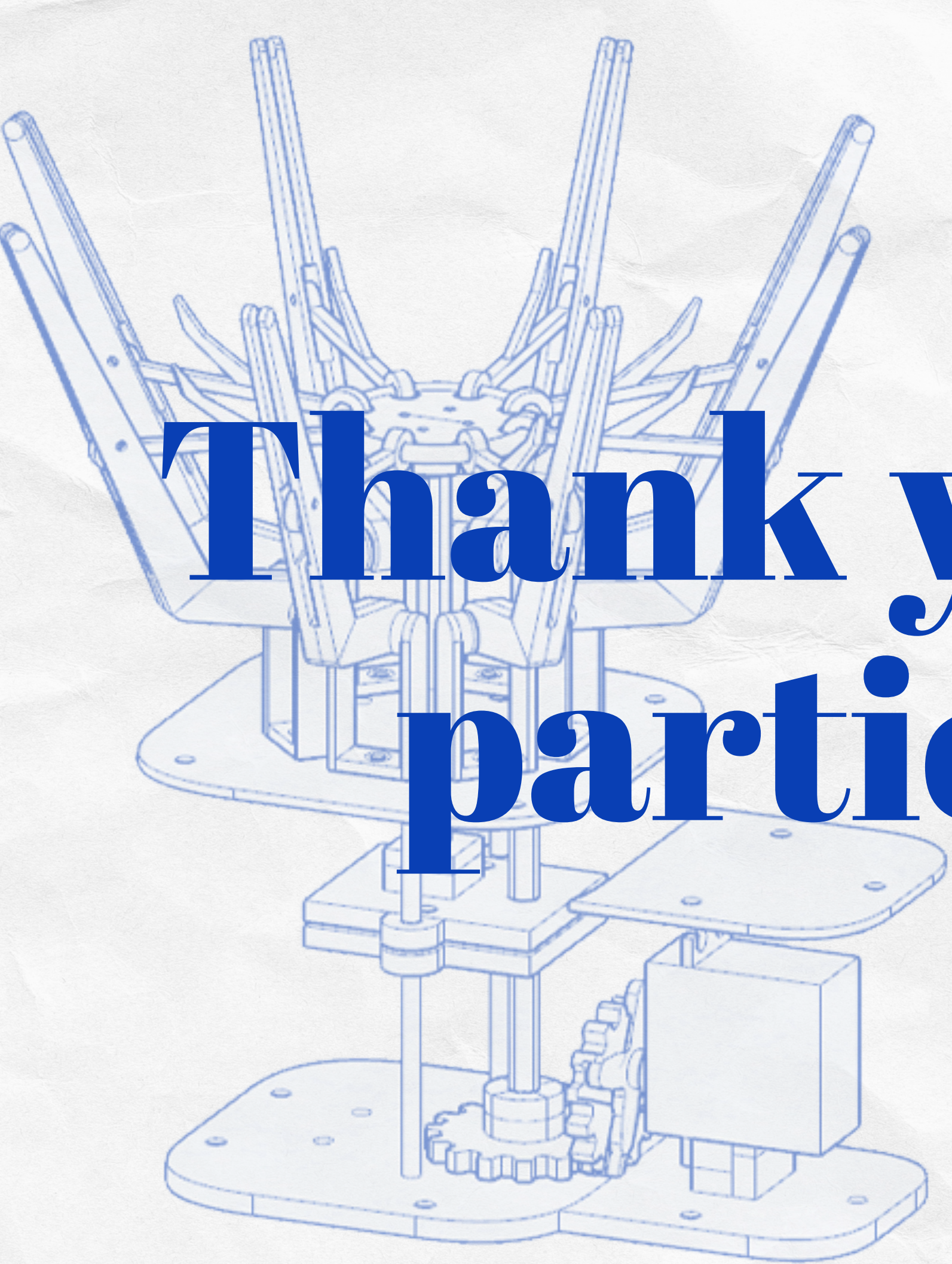


Let's Bloom Flower with Interaction!

Wire It Up!



Step	Component	Port	Connect To	Note
1	Arduino Nano	+5V	Breadboard Rail (+)	5V Power from Arduino Nano
		GND	Breadboard Rail (-)	Common ground for the system
2	BreadBoard	GND to GND	Breadboard Rail (-) to Another Breadboard Rail(-)	
3	Blue Button	Pin 1	D3	Logic signal input (Open/Reset)
		Pin 3	GND	Connects internal circuit to Ground
4	Orange Button	Pin 1	D4	Logic signal input (Close)
		Pin 3	GND	Connects internal circuit to Ground
5	Touch Sensor	S (Signal)	D5	Signal output to Arduino
		+ (VCC)	+5V	Power supply for the sensor
		- (GND)	GND	Ground for the sensor
6	Battery Pack	Black	GND	Essential for "Common Ground"
		Red	Another Breadboard Rail (+)	CRITICAL: Connect to Battery, NOT Nano
7	Servo Motor	Yellow	D7	PWM Control Signal
		Black	GND	Ground for the servo
		Red	Another Breadboard Rail (+)	6V Power
8	Arduino Nano	VIN	Another Breadboard Rail (+)	Connect After check all the connection!



Thank you for your participation!

3D Model: Max, Leonie
Coding: Wickey